

Forecasting 30-Day Realized Market Volatility Using Graph-Based Representations of Equity Return Structure

Bachelor's Thesis

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Abstract

This thesis examines whether networks built from U.S. equity returns can help forecast realized market volatility over the next 30 trading days. Many standard volatility approaches rely either on past volatility or on market-implied expectations, but these approaches may miss information contained in changing connections among individual stocks. When stocks become more synchronized and diversification weakens, these network changes may provide early signals of future volatility.

To study this question, the thesis implements a highly modular and configurable empirical pipeline and applies it to a canonical forecasting specification. The main analysis constructs a time series of graphs for a stable universe of 500 U.S. stocks, where firms are nodes and edge links are formed from rolling return correlations. These graphs are used to train a Graph Neural Network that produces one forecast per date for realized volatility over the next 30 trading days, measured from daily S&P 500 returns. Forecasts are evaluated in a strict chronological out-of-sample design against the VIX, a HAR-type historical volatility benchmark, and a placebo-graph ablation.

The results show that the graph-based model contains clear out-of-sample predictive information about future realized market volatility. Relative to the VIX, the model performs better on log-scale forecast errors in the full test sample and in several restricted-sample robustness exercises. The evidence is more mixed on full-sample variance-based QLIKE and on the question of whether the graph-based forecast contributes stable incremental information beyond the VIX in every sample. Diagnostic analysis further shows that the graph neural network performs especially well in low- and medium-volatility environments, while the VIX remains more informative in the highest-volatility regime. The placebo-graph comparison also indicates that the present implementation does not yet isolate a large marginal contribution from the exact baseline graph topology.

Overall, the thesis provides evidence that networks built from U.S. equity returns are useful for forecasting future market volatility. Its main contribution is not to show universal dominance over implied-volatility benchmarks, but to show that stock-return networks can generate useful forward-looking forecasts within a coherent out-of-sample framework. The thesis also contributes a modular and configurable empirical codebase that can support future experiments with alternative targets, graph designs, model specifications, and benchmark comparisons.

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1 Introduction

This thesis examines whether networks built from U.S. equity returns can help forecast realized market volatility over the next 30 trading days. The central idea is that future volatility may depend not only on past volatility or option-implied expectations, but also on how individual stocks are connected to one another. If stocks become more synchronized and diversification weakens before aggregate volatility fully rises, then the changing structure of the equity market may contain useful information about market turbulence.

Volatility forecasting matters because volatility affects asset pricing, portfolio allocation, risk management, derivatives pricing, and financial stability. When volatility rises unexpectedly, valuations become less stable, hedging costs change, leverage constraints tighten, and investors revise expectations. For this reason, academics and practitioners have long searched for reliable signals of near-term turbulence. Yet short-horizon volatility remains difficult to forecast consistently, especially out of sample and against strong market-based benchmarks.

Standard approaches rely on either historical market behavior or forward-looking option prices. Historical models, including ARCH, GARCH, and realized-volatility frameworks, use past return and volatility dynamics. Implied-volatility measures such as the VIX instead use option prices to summarize market expectations about future uncertainty. These approaches are valuable and difficult to outperform, but they mainly summarize volatility at the aggregate market level. They do not directly show how fragility is distributed across individual stocks, or whether the internal structure of the equity market is becoming more connected before realized volatility fully materializes.

Network methods offer a natural way to study this internal structure. Stocks do not move in isolation; they move together in patterns that change over time. In calmer periods, diversification may work relatively well because return comovements are less synchronized. During stress, correlations often rise and the market begins to behave more like a single connected system. If these structural changes contain early signs of weakening diversification, then stock-return networks may provide useful information for forecasting future volatility.

The choice of this topic was also shaped by my own methodological background. Most of my previous experience was with machine learning, neural networks, and convolutional neural networks, but I wanted to apply that knowledge to a finance problem where the structure of the data itself mattered. Equity markets seemed like a natural setting because stocks are not independent observations; they are linked through changing patterns of comovement, sector

exposure, investor behavior, and market-wide stress. I began to think of the stock market less as a flat list of returns and more as a living network whose structure expands, contracts, and reorganizes over time. Graph neural networks were attractive because they connected my interest in neural-network methods with a financial setting where relationships between observations are central rather than incidental. Volatility forecasting then became a meaningful target because it is economically important, difficult to predict, and connected to the VIX, a benchmark recognized by researchers and practitioners.

The literature gap follows from the distinction between describing market structure and forecasting with it. Financial-network studies show that markets become more connected during stress and that network measures can describe contagion, systemic risk, and fragility. However, showing that network structure changes during crises is not the same as showing that it can forecast future volatility before that volatility is realized. The gap addressed in this thesis is therefore whether stock-return networks contain useful out-of-sample predictive information for aggregate market volatility when tested chronologically and compared with a strong forward-looking benchmark such as the VIX. This makes the thesis academically relevant because it tests whether financial networks can move beyond description into prediction, and practically relevant because it asks whether market-structure information can provide an additional risk-monitoring signal alongside established volatility indicators.

The central research question is: Do networks built from U.S. equity returns contain out-of-sample predictive information for forecasting realized market volatility over the next 30 trading days? More specifically, the thesis asks whether a graph-based model can generate useful forecasts of future S&P 500 realized volatility, and whether those forecasts are competitive with or complementary to the VIX.

To answer this question, the thesis constructs graph snapshots for a stable universe of 500 U.S. stocks. Firms are represented as nodes, while links are formed from rolling return correlations. These graphs are used to train a graph neural network that produces one market-level volatility forecast per date. The target is forward 30-trading-day realized volatility, measured from daily S&P 500 returns. Forecasts are evaluated in a strict chronological out-of-sample framework against the VIX, a HAR-type historical-volatility benchmark, and a placebo-graph ablation.

The results show that the graph-based model contains clear out-of-sample predictive information about future realized volatility. Relative to the VIX, it performs better on log-scale forecast errors in the full test sample and in several restricted-sample robustness exercises. However, the evidence is more mixed on full-sample QLIKE and on stable incremental

information beyond the VIX. Diagnostics show that the model performs especially well in low- and medium-volatility environments, while the VIX remains more informative in the highest-volatility regime. The placebo comparison further suggests that the current implementation does not yet isolate a large marginal contribution from the exact baseline graph topology.

The thesis contributes by testing whether graph-based representations of stock-return dependence can be used for forecasting rather than only for describing market structure, and by evaluating that question in a demanding benchmark environment through VIX and HAR comparisons, placebo ablation, formal hypothesis testing, and calibration diagnostics. For academics, it provides evidence on both the forecasting value and limitations of graph-based representations of equity-market structure. For practitioners, it examines whether stock-return networks can provide an additional risk-monitoring signal alongside the VIX. A further practical contribution is the modular empirical pipeline, which can support future experiments with alternative graph designs, node features, forecasting targets, benchmarks, and descriptive outputs such as industry connectedness or changes in stock centrality.

The thesis is structured as follows. Section 2 reviews the literature on volatility forecasting, implied-volatility benchmarks, financial networks, and machine-learning approaches, and uses this literature to identify the research gap addressed in the study. Section 3 presents the empirical methodology, including the data sources, graph-construction procedure, model architecture, forecasting design, benchmark selection, and formal hypotheses. Section 4 reports the out-of-sample forecasting results, hypothesis tests, robustness checks, placebo comparison, and calibration diagnostics. Section 5 discusses the interpretation of the findings, their practical implications for risk monitoring, and the main limitations of the empirical design. Section 6 concludes by summarizing the contribution of the thesis and outlining directions for future research.

2 Literature Review

2.1 Foundations of Volatility Modelling

2.1.1 *Conditional Volatility Models: ARCH and GARCH*

The volatility-forecasting literature begins from the empirical fact that financial returns do not have constant variance over time. Large price movements tend to cluster, so periods of market calm and market turbulence arrive in persistent sequences rather than as independent events. Engle formalized this insight through the ARCH model, in which conditional variance depends on past squared shocks and volatility becomes an object of direct modelling rather than a residual nuisance (Engle, 1982). Bollerslev then generalized this framework through GARCH, allowing current variance to depend on both past innovations and past conditional variance. This made volatility persistence more parsimonious to estimate and established GARCH-type models as central benchmarks in empirical finance (Bollerslev, 1986).

The importance of ARCH and GARCH for this thesis is not that they provide the final forecasting model, but that they define the historical-volatility tradition against which later approaches are understood. They show that volatility contains predictable structure in its own past. At the same time, their basic forms are limited because they rely on lagged market information and impose relatively simple dynamics on a process that is asymmetric, persistent, and state dependent (Poon & Granger, 2003).

2.1.2 *Extensions of GARCH and Long-Memory Models*

Subsequent work extended the basic GARCH framework to better capture empirical features of financial returns. One important limitation is symmetry: in standard GARCH, positive and negative shocks of equal size have the same effect on future volatility, even though equity-market volatility often reacts more strongly to negative news. EGARCH and GJR-GARCH address this issue by allowing volatility to respond asymmetrically to shocks, thereby capturing leverage effects more directly (Nelson, 1991; Glosten, Jagannathan, & Runkle, 1993).

A second limitation concerns persistence. Standard GARCH implies that shocks to volatility decay relatively quickly, whereas financial volatility often displays slower and more durable dependence. Ding, Granger, and Engle show that absolute returns exhibit strong persistence, motivating long-memory approaches and reinforcing the idea that volatility dynamics extend beyond short autoregressive adjustment (Ding, Granger, & Engle, 1993). These extensions matter because they show that even within historical models, the forecasting problem is richer than simple lag dependence. However, they remain fundamentally backward-

looking: they improve the modelling of past volatility dynamics but do not directly incorporate forward-looking market expectations or cross-sectional market structure.

2.1.3 Realized Volatility Frameworks

A second major development came from realized-volatility frameworks. Instead of treating volatility as latent and inferring it only from daily returns, this literature constructs ex post measures of variation by aggregating higher-frequency return information. Barndorff-Nielsen and Shephard link realized volatility to quadratic variation, giving it a formal statistical foundation as a measure of return variability (Barndorff-Nielsen & Shephard, 2002). Andersen, Bollerslev, Diebold, and Labys then show how realized volatility can be used in a coherent framework for measuring and forecasting market uncertainty across horizons (Andersen, Bollerslev, Diebold, & Labys, 2003).

The HAR-RV model is especially relevant for this thesis because it provides a simple and influential way to summarize persistence across daily, weekly, and monthly volatility components (Corsi, 2009). Later models such as HEAVY and Realized GARCH further integrate realized measures into conditional-volatility forecasting by linking high-frequency variation to future volatility dynamics (Shephard & Sheppard, 2010; Hansen, Huang, & Shek, 2012). These models strengthen the historical-volatility benchmark, but they also clarify its limitation. Realized volatility is informative because markets are persistent, yet it is still constructed from past market activity. This motivates the next section's focus on implied volatility and the VIX, which introduce forward-looking information from option prices.

2.2 Implied Volatility and the VIX as a Forward-Looking Benchmark

2.2.1 Concept and Construction of Implied Volatility

Whereas realized volatility is constructed from past returns, implied volatility is extracted from current option prices and therefore provides a forward-looking market-based measure of expected uncertainty. In the Black-Scholes framework, an option price depends on observable inputs such as the underlying asset price, strike price, time to maturity, and risk-free rate, together with an unobserved volatility parameter. Implied volatility is the value of that parameter which makes the model-implied option price equal to the observed market price (Black & Scholes, 1973).

This matters because implied volatility converts derivative prices into a standardized signal about future variation. Unlike ARCH, GARCH, and realized-volatility approaches, which rely on historical return information, implied volatility reflects information embedded in option

markets at the time the forecast is made. It is therefore a natural forward-looking benchmark in volatility forecasting, while still not being a pure or mechanically unbiased forecast of future realized volatility (Poon & Granger, 2003).

2.2.2 The VIX as a Measure of Expected Market Volatility

The practical importance of implied volatility is most visible in the VIX, which has become the standard market benchmark for short-horizon U.S. equity uncertainty. The VIX summarizes expected S&P 500 volatility over approximately the next thirty calendar days using a broad range of out-of-the-money index options. In its modern form, it is designed as a model-free measure of expected variance and is expressed in annualized volatility terms, making it broader than the implied volatility of any single option contract (Cboe Exchange, Inc., 2019).

The VIX is also widely interpreted as a market-stress indicator. When uncertainty rises and investors demand more protection, option prices tend to increase, and the VIX rises with them. Although it reflects risk-neutral rather than purely physical expectations, its visibility and economic meaning make it one of the most important reference points for both academic forecasting studies and practical risk monitoring (Whaley, 2000).

2.2.3 Empirical Performance of the VIX in Forecasting

The empirical literature generally finds that implied-volatility measures contain substantial information about future realized volatility and often outperform purely historical time-series models, especially at short horizons. This is intuitive because option prices can incorporate new information and changing risk perceptions more quickly than models based only on lagged returns. Survey evidence therefore places implied volatility among the strongest single predictors of subsequent volatility, while also emphasizing that performance depends on the horizon, benchmark specification, and realized-volatility proxy used for evaluation (Poon & Granger, 2003).

More focused evidence reaches a similar conclusion. Blair, Poon, and Taylor show that option-implied volatility for the S&P 100 contains incremental forecasting information beyond historical returns and high-frequency return measures (Blair, Poon, & Taylor, 2001). For this thesis, the implication is clear: the VIX is not a weak benchmark included for convenience, but a demanding forward-looking comparator. A graph-based model that performs well against it must therefore contain information that is at least partly competitive with information already priced in option markets.

2.2.4 Limitations of Implied Volatility Measures

Despite their forecasting value, implied-volatility measures have important limitations. Because they are extracted from option prices, they reflect not only expected future variance but also volatility risk premia, market liquidity, hedging demand, and other derivative-market conditions. They should therefore be treated as market-implied indicators rather than pure statistical expectations of realized volatility (Poon & Granger, 2003). This distinction is important for the present thesis because the VIX is used as a benchmark, not as a direct measure of the true conditional expectation of future volatility.

A second limitation is structural. Even model-free measures such as the modern VIX summarize expectations about aggregate index-level variance. They do not reveal how fragility is distributed across individual firms, whether dependence among stocks is becoming more concentrated, or whether the internal topology of the equity market is changing before volatility is fully realized. Earlier disagreement in the implied-volatility literature also shows that results can depend on sample design and measurement choices rather than on the simple superiority or failure of implied volatility itself (Christensen & Prabhala, 1998). For this thesis, this limitation creates the opening for network-based forecasting: the VIX is a strong forward-looking benchmark, but it does not directly measure the cross-sectional structure through which market stress may build and spread (Jiang & Tian, 2005).

2.3 Financial Networks and Market Structure

2.3.1 Correlation-Based Networks in Financial Markets

Financial-network analysis responds to the limits of aggregate indicators such as the VIX by asking whether the market's cross-sectional structure contains information about fragility. It treats assets as nodes and statistical relationships between returns as links, converting dependence into a graph representation of market organization. Mantegna provided the seminal formulation by mapping correlations into distances and extracting a minimum spanning tree, showing that the resulting structure recovered economically meaningful groupings of firms (Mantegna, 1999).

Graph-construction choices are central because different topologies embody different assumptions about dependence. Minimum spanning trees and related filters retain a sparse backbone of strong relationships, while threshold and top-k graphs preserve links that meet a strength or ranking rule, making them useful for stable and computationally manageable neighborhood structures. Partial-correlation networks attempt to isolate more direct relationships after accounting for common market-wide variation, while spillover networks

focus on directional transmission. These alternatives are not interchangeable: the estimated network depends on the dependence measure, filtering rule, window length, and sparsification procedure used.

For this thesis, correlation-based networks provide a tractable way to move from aggregate expected volatility to the internal architecture of the equity market. They do not identify causal transmission, and this thesis does not treat them as evidence of contagion. Their value is predictive and representational: they allow the empirical framework to ask whether stock dependence becomes more concentrated, clustered, or synchronized before stress is fully visible in market-wide volatility. This makes graph construction a form of network-based feature engineering rather than a causal model of financial propagation.

2.3.2 Network Topology and Market Dynamics

Once correlations are represented as networks, the literature shifts from individual links to the market's overall shape. Measures such as centrality, clustering, path length, and network distance describe whether the market is dispersed, hierarchically organized, or compressed around dominant nodes. This matters because topology captures system-level structure not visible from isolated pairwise correlations.

A central finding is that network topology changes systematically with market conditions. Onnela et al. show that during Black Monday the minimum spanning tree contracts, meaning that distances between assets shrink as correlations intensify and the system becomes more synchronized (Onnela, Chakraborti, Kaski, & Kertész, 2003). Sandoval and de Paula Franca similarly show that crisis periods are associated with sharply rising correlations across financial markets, making the system behave more like a single connected structure (Sandoval & De Paula Franca, 2012).

For this thesis, this literature turns changing co-movement into measurable structural information. If topology becomes more concentrated, compressed, or synchronized before severe stress, those changes may complement aggregate indicators such as the VIX. However, topology should be interpreted carefully: a more connected network can signal weakening diversification, but does not by itself prove that shocks are causally transmitted across firms.

2.3.3 Financial Contagion and Systemic Risk

The literature on financial contagion and systemic risk extends network analysis from co-movement to propagation. Whereas correlation-based equity networks show which assets move together, contagion models ask whether distress at one node can be transmitted through the system and generate losses elsewhere. In this setting, systemic risk refers to the possibility that

shocks are amplified by the pattern of interconnections rather than remaining isolated. Allen and Gale provide one of the earliest formal treatments, showing that interconnected networks can improve risk sharing in normal times while still leaving the system vulnerable when shocks hit particular structures (Allen & Gale, 2000).

Subsequent work develops this robustness-fragility trade-off. Gai and Kapadia show that greater connectivity can reduce the probability that contagion begins but increase the scale of cascades once failures occur (Gai & Kapadia, 2010). Acemoglu, Ozdaglar, and Tahbaz-Salehi similarly argue that dense networks can absorb small shocks while amplifying sufficiently large ones through the same interconnections that ordinarily support stability (Acemoglu, Ozdaglar, & Tahbaz-Salehi, 2015). Battiston et al. reinforce this point by showing that systemic importance depends not only on size but also on position within the network (Battiston, Puliga, Kaushik, Tasca, & Caldarelli, 2012).

This literature matters for the present thesis mainly as conceptual motivation. It explains why market structure may matter for fragility, but the empirical exercise here is narrower. The thesis does not estimate causal contagion channels or identify shock transmission mechanisms. Instead, it uses network structure as a predictive representation of changing equity-market dependence.

2.3.4 Networks as Indicators of Market Stress

As the network literature developed, researchers increasingly treated network measures not only as descriptions of market organization but also as indicators of mounting stress. Crises are often associated with tighter co-movement, shorter network distances, and greater concentration of influence, so changes in these statistics may reveal weakening diversification before stress is fully visible in aggregate outcomes. Evidence from correlation-based market networks supports this view by showing that crisis periods involve stronger synchronization and more compact market structures (Onnela, Chakraborti, Kaski, & Kertész, 2003; Sandoval & De Paula Franca, 2012).

Later work developed more explicit monitoring frameworks. Billio et al. construct connectedness measures for the finance and insurance sectors and argue that rising interconnectedness is informative about systemic risk (Billio, Getmansky, Lo, & Pelizzon, 2012). Diebold and Yilmaz similarly show that variance decompositions can be interpreted as weighted, directed networks for measuring time-varying connectedness across financial firms (Diebold & Yilmaz, 2014). The key point is that linkage intensity and direction can be tracked as state variables summarizing system-wide vulnerability.

For this thesis, these studies bridge market structure and volatility forecasting. They suggest that network measures can summarize fragility not directly visible in aggregate indicators. However, most evidence remains descriptive, contemporaneous, or diagnostic. Showing that network structure changes during crises is not the same as showing that it predicts future volatility out of sample. That distinction motivates the next section, which turns from networks as stress representations to networks as forecasting inputs.

2.4 Predictive Use of Financial Networks

2.4.1 Descriptive to Predictive Network Analysis

The previous section shows that financial networks can describe market structure and identify periods of stress. The next step is to ask whether those same structures can be used predictively. This distinction is important. A network statistic that rises during a crisis is not necessarily useful for forecasting if it only moves contemporaneously with realized volatility. Predictive network analysis therefore requires a stricter question: whether information available at date t about connectedness, synchronization, or topology helps forecast outcomes at a later date.

Billio et al. and Diebold and Yilmaz helped move the literature in this direction by treating connectedness as a time-varying state variable rather than a static description of market structure (Billio, Getmansky, Lo, & Pelizzon, 2012; Diebold & Yilmaz, On the network topology of variance decompositions: Measuring the connectedness of financial firms, 2014). Once networks are measured repeatedly through time, they can be used not only to summarize current vulnerability but also to test whether changes in vulnerability precede future market stress. Magner et al. make this predictive turn especially clear by showing that financial-network measures can improve realized-volatility forecasts relative to specifications that ignore cross-asset dependence (Magner, Lavin, Valle, & Hardy, 2020). For this thesis, this shift is central: the empirical question is not whether networks describe crises after they occur, but whether dated equity-network information contains usable out-of-sample predictive content for future volatility.

2.4.2 Network Features as Predictors of Volatility and Risk

In much of the predictive literature, the network is not used directly as a raw object. Instead, researchers transform it into features that summarize economically meaningful dimensions of market structure. One group captures synchronization. Measures such as minimum spanning tree length, average network distance, or filtered-network compactness indicate whether assets

are moving more tightly together. A shorter or more compressed network is often interpreted as weaker diversification and stronger market-wide co-movement. Magner et al. use this logic to show that network-based measures can improve volatility forecasts across several stock markets (Magner, Lavin, Valle, & Hardy, 2020).

A second group of features focuses on connectedness and spillovers. Diebold and Yilmaz show that forecast-error variance decompositions can be interpreted as directional networks, allowing total and directional spillovers to be tracked over time (Diebold & Yilmaz, 2012). Ahelegbey and Giudici extend this idea through NetVIX, which combines volatility and network interconnectedness into a measure of global market risk (Ahelegbey & Giudici, 2022). These studies show that network feature engineering can convert high-dimensional dependence into tractable predictors such as synchronization, centrality, connectedness, concentration, or spillover intensity.

This feature-engineering perspective is important here. The graph snapshots used later are constructed forecasting inputs, not neutral objects. The choice of dependence measure, rolling window, sparsification rule, and node information determines what form of market structure the model can learn from. Unlike much of the earlier predictive literature, this thesis does not only feed handcrafted network statistics into a conventional regression. It uses graph-based representations that allow a graph neural network to learn from node-level information and relational structure among firms. Even so, graph construction remains feature engineering because it defines the model's information set.

2.4.3 Challenges in Using Network Data for Forecasting

Despite their promise, network-based forecasting approaches face important methodological challenges. The first is construction risk. Network measures are not directly observed variables; they are produced by choices about the return window, dependence metric, filtering rule, and graph topology. Different choices can produce different networks and therefore different forecasting signals (Tumminello, Aste, Di Matteo, & Mantegna, 2005). A correlation network, partial-correlation network, minimum spanning tree, and spillover network may all describe connectedness, but they do not measure the same object.

A second challenge is that there is no single agreed representation of financial connectedness. Billio et al. measure interconnectedness through principal components and Granger-causality networks, whereas Diebold and Yilmaz derive directed connectedness from forecast-error variance decompositions (Billio, Getmansky, Lo, & Pelizzon, 2012; Diebold &

Yilmaz, 2014). These approaches capture related but distinct forms of dependence, which makes findings harder to compare and increases the need to justify the chosen topology.

A third challenge is stability and generalizability. Magnier et al. show that network-based measures can improve volatility forecasts, but the gains are not uniform across all markets in their sample (Magnier, Lavin, Valle, & Hardy, 2020). This cautions against treating network features as automatically superior predictors. Forecasting gains may come from the representation, the model architecture, historical volatility embedded in the features, or sample-specific crisis behavior. Credible network forecasting therefore requires chronological out-of-sample evaluation, strong benchmarks, and diagnostics that test whether performance survives alternative samples or graph constructions.

A final challenge is interpretation. Network forecasts should not automatically be read as causal evidence that shocks travel through estimated links. In this thesis, the network is a predictive representation of cross-sectional dependence, not a causal map of contagion. The relevant question is narrower but empirically cleaner: whether graph-based representations of equity-return structure contain useful information for forecasting future realized volatility out of sample and against demanding benchmarks.

2.5 Machine Learning in Volatility Forecasting

2.5.1 Motivation for Machine Learning Approaches

The growing use of machine learning in volatility forecasting reflects a change in the forecasting problem itself. Traditional econometric models such as GARCH, HAR, and implied-volatility benchmarks remain essential because they are interpretable, parsimonious, and grounded in financial theory. However, once the information set expands to include realized measures, macro-financial variables, and network-based representations of market structure, the problem becomes more high-dimensional and non-linear. In this setting, machine learning is useful not because it automatically replaces econometric models, but because it can process richer predictor sets and capture interactions that are difficult to impose through a small parametric specification (Gu, Kelly, & Xiu, 2020).

This motivation is directly relevant to volatility forecasting. Zhu et al. show that machine learning methods can exploit broader information sets to improve realized-volatility forecasts relative to conventional linear benchmarks (Zhu, Bai, He, & Liu, 2023). For this thesis, the role of machine learning is therefore practical rather than ideological. The question is whether

flexible models can extract useful predictive structure once volatility forecasting is combined with network-based feature engineering.

2.5.2 Tree-Based and Regularized Models

Tree-based and regularized models occupy an intermediate position between classical econometric models and neural networks. Tree-based methods can capture non-linearities and interactions without requiring the researcher to specify the functional form in advance. Regularized models, such as lasso, ridge, and elastic net, retain a more transparent linear structure while shrinking coefficients and reducing overfitting when the predictor set is large or highly correlated.

These methods are relevant because volatility forecasting often involves many candidate predictors, including lagged realized measures, market variables, and network-derived features. Dai et al. show that shrinkage methods can improve stock-return volatility forecasts relative to autoregressive and competing benchmarks (Dai, Li, & Yang, 2022). Their contribution is also a useful reminder for the present thesis: richer information sets are valuable only if model complexity is controlled. This makes validation and out-of-sample evaluation central to any machine-learning forecasting design.

2.5.3 Deep Learning and Graph Neural Networks

Deep learning extends this logic by using architectures designed to learn non-linear representations directly from data. In volatility forecasting, many applications rely on recurrent architectures such as LSTM networks because volatility is persistent and path-dependent. Kim and Won show that hybrid LSTM models with GARCH-type inputs can outperform single econometric models and simpler neural-network alternatives in forecasting volatility (Kim & Won, 2018). The broader lesson is that deep learning is most useful when its architecture matches the structure of the forecasting problem.

Graph neural networks are especially relevant when the predictors are relational rather than purely tabular or sequential. A standard machine-learning model usually receives a vector of engineered predictors, while a GNN can process node-level information together with links among observations. In this thesis, firms are nodes, return dependence defines edges, and the forecasting target is market-level volatility. A graph model is therefore a natural architecture because it can aggregate information from stocks while preserving the cross-sectional structure of their relationships. GraphSAGE is particularly suitable for this setting because it learns node representations by aggregating information from local neighborhoods, making it appropriate for large graph-structured datasets (Hamilton, Ying, & Leskovec, 2017).

This does not mean that a GNN is automatically optimal. Its value depends on whether the graph construction, node features, and validation design allow the model to learn useful predictive structure rather than noise. The methodological question is therefore whether a graph-based neural architecture can convert equity-market dependence into reliable out-of-sample volatility forecasts.

2.5.4 Comparative Evidence vs Traditional Models

The comparative evidence cautions against treating machine learning as a universal replacement for traditional volatility models. Forecasting performance depends on the target variable, horizon, information set, benchmark, and evaluation design rather than on the unconditional superiority of one model class (Poon & Granger, 2003). More recent studies reach a similar conclusion: machine-learning methods can outperform classical linear specifications, especially in short-horizon settings with richer data, but the gains are not universal and often depend on design choices (Zhu, Bai, He, & Liu, 2023; Díaz, Hansen, & Cabrera, 2024).

For this thesis, the implication is that machine learning should be treated as a conditional forecasting tool, not as an automatic improvement. Traditional models and implied-volatility measures remain important because they are difficult benchmarks to beat. The relevant question is narrower: whether a GNN can extract additional predictive value from graph-based representations of equity-return structure when tested chronologically, compared with strong benchmarks, and interpreted as evidence of predictability rather than causality.

2.6 Synthesis, Theoretical Framework, and Research Gap

2.6.1 Integration of the Literature

The literature reviewed above points to a clear tension in volatility forecasting. Historical models such as ARCH, GARCH, HAR-RV, and related extensions show that volatility is persistent and partly forecastable from its own past. Implied-volatility measures, especially the VIX, add forward-looking information by incorporating option-market expectations in real time. Together, these literatures create a demanding benchmark environment: any new approach must be assessed against both historical models and a market-based measure that is already difficult to outperform.

The financial-network literature adds a different perspective. Aggregate indicators such as realized volatility or the VIX summarize market-wide uncertainty, but not how fragility is organized across firms. Correlation-based networks, connectedness measures, and topological indicators show that during stress, assets become more synchronized, network distances

contract, and dependence patterns concentrate. Predictive network studies further suggest that some structural features may contain information about future volatility and risk, rather than only describing stress after it occurs.

Machine learning becomes relevant once these strands are combined. A forecasting design using dated equity-network information is no longer based only on a small number of lagged aggregate variables. It involves a high-dimensional and potentially non-linear representation of firm-level attributes and inter-firm dependence. Graph-based learning methods are therefore natural because they process node information together with relationships among nodes. In this sense, the thesis uses a hybrid econometric-ML forecasting framework with network-based feature engineering: historical and implied volatility define the benchmark environment, while graph construction and graph learning define the alternative information set being tested.

2.6.2 Identified Gaps

Three gaps follow from this synthesis. First, much of the financial-network literature remains descriptive, contemporaneous, or diagnostic. It shows that network topology changes during crises and that connectedness rises in turbulent periods, but this is not the same as showing that networks can forecast future volatility out of sample. A variable that moves with crises *ex post* is not necessarily useful *ex ante*. The relevant unresolved question is whether equity-network information available at date t contains predictive content for realized market volatility over a future horizon.

Second, the predictive literature often relies on handcrafted network statistics rather than models that operate on graph-structured representations themselves. Summary measures such as connectedness, centrality, or network distance are useful because they are interpretable, but they also compress the network before estimation. This may discard information contained in the broader cross-sectional pattern of dependence. Graph-based learning offers one way to test whether richer market-structure representations contain forecasting value beyond a small set of preselected indicators.

Third, the credibility of network-based volatility forecasting depends on how it is evaluated. A model may appear useful because of the graph representation, the machine-learning architecture, historical volatility information embedded in the features, or sample-specific crisis behavior. It may also perform well on average without adding stable information beyond implied volatility. For that reason, the literature leaves open a benchmark and validation gap: graph-based forecasts should be tested chronologically, compared with strong alternatives such as the VIX and historical-volatility benchmarks, and examined through diagnostics that

separate predictive content, relative accuracy, incremental information, robustness, and calibration.

These gaps define the space in which the present thesis operates. The thesis does not ask whether networks prove causal contagion, nor whether volatility is forecastable in general. Its question is narrower: whether graph-based representations of U.S. equity-return structure contain useful out-of-sample predictive information for forward realized market volatility when evaluated against demanding benchmarks.

2.6.3 Contribution of This Thesis

This thesis contributes by bringing the volatility-forecasting, implied-volatility, financial-network, and machine-learning literatures into a single empirical framework. It constructs dated graph snapshots from a stable universe of U.S. equities, uses them to train a graph neural network, and evaluates forecasts of forward 30-trading-day realized market volatility in a strict chronological out-of-sample design. The benchmark set includes the VIX, a HAR-type historical-volatility model, and a placebo-graph ablation.

The contribution is threefold. First, the thesis moves financial-network analysis from description toward forecasting by asking whether dated cross-sectional dependence can predict subsequent aggregate volatility. Second, it evaluates this question in a demanding benchmark environment. Treating the VIX as the main comparator makes the test economically meaningful because the graph-based model is judged against a market-implied measure that already aggregates option-market information. Third, the thesis addresses model-risk concerns by combining forecast-loss comparisons with formal hypothesis tests, placebo ablation, restricted-sample robustness checks, and calibration diagnostics.

The thesis should therefore be understood as a test of predictability rather than causality. It does not identify a structural contagion mechanism or claim that estimated graph links are causal transmission channels. Instead, it asks whether cross-sectional equity-return dependence can be engineered into graph-based forecasting inputs that generate useful forward-looking signals. Framed this way, the thesis contributes substantively by testing the forecasting value of equity-market structure, and methodologically by combining graph construction, graph learning, benchmark comparison, and robustness diagnostics in a coherent volatility-forecasting design.

3 Methodology

3.1 Analytical Framework and Forecasting Design

This thesis uses a hybrid econometric–machine-learning forecasting framework with network-based feature engineering. The econometric component defines the forecasting target, the benchmark structure, the out-of-sample evaluation design, and the formal hypothesis tests. The machine-learning component uses graph neural networks to transform cross-sectional equity-return information into a market-level volatility forecast. The network feature-engineering component lies between these two parts: daily firm-level returns are converted into dated graph snapshots, where firms are represented as nodes and rolling return dependence defines the links between them.

The purpose of this framework is not to estimate a structural model of financial contagion. The graph is used as a predictive representation of market structure rather than as a causal map of shock transmission. This distinction is central to the empirical design. The thesis asks whether information available at date t , including the recent behavior and dependence structure of U.S. equities, helps forecast realized market volatility over the future interval $t + 1, \dots, t + 30$. It does not claim that estimated graph links identify causal transmission channels between firms. The empirical question is therefore one of predictability, not causality.

This section describes the forecasting target, data inputs, graph-construction procedure, model architecture, validation strategy, robustness design, and formal hypothesis tests. Further implementation details, including raw input schemas, canonical configuration settings, supplementary diagnostics, and alternative model configurations, are reported in the appendices.

3.1.1 Forecasting Target

The forecasting target is forward 30-trading-day realized market volatility for U.S. equities, constructed from daily S&P 500 returns. For each feature date t , the target is measured over the future window $t + 1, \dots, t + 30$. This horizon is chosen to approximate the one-month volatility horizon commonly associated with the VIX while preserving a trading-day definition that is directly measurable from realized index returns.

Forward realized volatility is annualized to 252 trading days and scaled by 100, so that values can be interpreted in percentage volatility terms:

$$RV_{30,t}^{fwd} = 100 \sqrt{\frac{252}{30} \sum_{j=1}^{30} r_{m,t+j}^2}$$

where $r_{m,t+j}$ denotes the daily S&P 500 return on future day $t + j$. The model is trained on the logarithm of realized volatility:

$$y = \log(RV_{30,t}^{fwd})$$

The log transformation reduces skewness, limits the influence of extreme volatility observations, and makes proportional forecast errors more central to estimation. Model forecasts can then be transformed back into volatility-level terms by exponentiation when evaluation outside log space is required. This allows the thesis to evaluate forecasts both in the model's training space and in the natural scale of volatility.

3.1.2 Benchmark Forecast

The main benchmark forecast is the VIX observed at feature date t . The VIX is used because it is the most widely recognized market-implied measure of expected short-horizon U.S. equity volatility and represents a demanding practical benchmark. Unlike the graph-based model, which uses only information derived from equity returns and their dependence structure, the VIX incorporates information from option prices and therefore reflects market expectations, volatility risk premia, hedging demand, and derivative-market conditions.

In level form, the benchmark forecast is:

$$\widehat{RV}_{30,t}^{VIX} = VIX_t$$

and in logarithmic form:

$$\hat{y}_t^{VIX} = \log(VIX_t)$$

The VIX is treated strictly as a benchmark rather than as an input to the graph neural network. This is important because the central empirical question is whether equity-derived network information can generate useful forward-looking forecasts without directly using implied-volatility information. The comparison is not a perfect horizon match, since the VIX refers approximately to 30 calendar days whereas the target is constructed over 30 trading days. Nevertheless, it remains the closest and most economically meaningful market-based comparator for the forecasting problem studied in this thesis.

3.1.3 Information Set and No-Look-Ahead Design

The empirical design imposes a strict no-look-ahead structure. For each feature date t , all predictors are constructed only from information available at or before t . Graph snapshots are built from a rolling historical window ending at t , and the target is realized volatility over the future interval $t + 1, \dots, t + 30$. This separation ensures that the dependent variable is not observable when the predictors are formed.

The chronological split reinforces this design. The training sample covers dates up to 31 December 2014, the validation sample covers 2015 to 2018, and the test sample covers 2019 to 2024. Although the raw inputs and configured test boundary extend to 31 December 2024, the final usable feature date in the canonical 30-trading-day target construction is 15 November 2024, because later feature dates do not have a complete forward 30-trading-day realized-volatility label. The model is estimated only on the training period, tuned using the validation period, and evaluated formally on the held-out test period. Feature scaling is also estimated using the training sample only and then applied unchanged to validation and test observations. These steps are designed to prevent leakage from future observations into either feature construction, model selection, or final evaluation.

The resulting framework is pseudo-out-of-sample rather than purely in-sample. This is essential because the thesis evaluates forecasting performance, not retrospective fit. The relevant question is whether the graph-based model produces useful predictions when tested chronologically against benchmarks that were not used to train the model.

3.2 Data and Graph Construction

3.2.1 Data Sources, Sample Period, and Universe Requirements

The empirical dataset combines three main inputs. First, a CRSP-based daily equity panel provides the stock-level returns, identifiers, prices, and shares outstanding used to construct the equity universe, node features, and graph edges. Second, daily S&P 500 returns are used to construct the forward realized-volatility target. Third, daily VIX observations are merged by date for benchmark evaluation only. The VIX is not used as a model input.

The underlying sample spans 3 January 2000 to 31 December 2024. Because graph construction requires a backward-looking rolling window and target construction requires a complete forward 30-trading-day window, the effective modeling sample begins after sufficient historical observations are available and excludes final dates without a complete target horizon.

The stock universe is designed to preserve a stable graph size while avoiding an ex post fixed-constituent design. At each feature date, the empirical pipeline maintains a universe of 500 eligible U.S. stocks selected by market capitalization. Eligibility is evaluated using information available within the rolling historical window, and firms that become ineligible are replaced by the next eligible stocks by market capitalization. This fixed-replace design maintains comparable graph dimensions over time while allowing the cross-section to evolve. The raw input schemas and required variables are summarized in Appendix A. In the canonical

implementation, stock-level rolling features and correlation-based graph edges are constructed from CRSP RETX, carried in the pipeline as `ret_used`. Delisting fields are retained in the raw schema for transparency and diagnostics, but delisting returns are not incorporated into the canonical return adjustment.

3.2.2 Construction of the Stable 500-Stock Universe

At the first valid feature date, all stocks are screened using the historical window ending at that date. A stock is eligible if it satisfies the minimum observation requirement within the window, set in the canonical specification at 80 percent of trading days. Among eligible stocks, the 500 largest by market capitalization are selected to form the initial graph universe.

For each subsequent feature date, eligibility is re-evaluated using only information available up to that date. Firms already in the universe are retained if they continue to satisfy the data-availability requirement. If a firm becomes ineligible, it is replaced by the largest eligible firm not already included. This fixed-replace procedure is applied at the feature-date level in the canonical implementation. The pipeline is designed to maintain a 500-stock graph whenever sufficient eligible securities are available, while avoiding an ex post fixed-constituent design. Market capitalization is computed from CRSP price and shares outstanding data as provided in the raw extract.

This design is especially important for a graph neural network because changes in graph size can complicate comparison across dates. Maintaining a constant number of nodes improves computational stability and makes graph snapshots more comparable, while the replacement rule prevents the analysis from relying only on firms that survive until the end of the sample.

3.2.3 Network Feature Engineering and Graph Topology Choice

The graph-construction step is a form of feature engineering. The network is not directly observed in the data; it is created by choosing a dependence measure, a rolling window, a sparsification rule, and a representation of firm-level information. These choices determine what type of market structure the model can learn from. They also create model risk: forecasting gains may arise from the node features, the graph topology, the GNN architecture, historical volatility information embedded in the inputs, or sample-specific market conditions. For that reason, the graph design must be justified rather than treated as a neutral preprocessing step.

Several graph-construction methods are possible in financial-network applications. Pearson correlation networks measure linear return dependence and are simple to interpret, but they may be sensitive to outliers. Spearman rank correlation networks measure monotonic co-movement and are somewhat more robust to extreme return observations. Partial-correlation

networks attempt to isolate more direct relationships after controlling for common variation, but they are more demanding to estimate reliably in high-dimensional rolling windows. Minimum spanning trees and related filtering methods produce sparse backbones of the strongest relationships, but they may discard useful local neighborhood information. Threshold and top-k graphs preserve stronger links while keeping graph density manageable. Spillover networks, such as those based on variance decompositions, can capture directional dependence but require a separate time-series modeling layer and are less directly aligned with the current equity-return graph setup.

The canonical specification uses Spearman rank correlation with a top-k sparsification rule. This choice is motivated by three considerations. First, Spearman correlation provides a robust dependence measure for rolling equity returns, where extreme observations can strongly affect linear correlation estimates. Second, top-k filtering gives each stock a local neighborhood of its strongest contemporaneous return-dependence relationships, which is well aligned with the neighborhood-aggregation logic of GraphSAGE. Third, the resulting graph is sparse enough to be computationally tractable while still preserving cross-sectional structure beyond a small number of aggregate network statistics.

This does not imply that the chosen topology is uniquely optimal. Partial-correlation networks, minimum spanning trees, alternative correlation measures, and different filtering rules are all plausible alternatives. The thesis therefore treats the baseline graph as a transparent and scalable canonical specification rather than as a final structural representation of the equity market. Appendix E reports alternative tuned configurations using different graph-construction windows, dependence measures, top-k settings, and architecture choices. These checks do not exhaust all possible topologies, but they help assess whether the main findings are specific to the baseline configuration.

3.2.4 Graph Snapshots, Node Features, and Edge Formation

For each feature date t , a graph snapshot $G_t = (V_t, E_t)$ is constructed. The node set V_t consists of the 500 stocks selected by the universe-construction procedure. Each node is assigned two baseline features: rolling mean return and rolling return volatility, both computed over the historical window ending at t . These features provide a compact summary of recent firm-level behavior and are intentionally parsimonious to keep the baseline model interpretable.

Edges are constructed from pairwise return dependence over the same rolling window. In the canonical specification, Spearman rank correlations are computed for all stocks in the date-specific universe. For each node, the $k = 10$ strongest counterparties by absolute correlation

are retained. The graph is then symmetrized to produce an undirected sparse graph. Edge weights preserve the signed correlation values, although the canonical GraphSAGE implementation used in the thesis relies on graph connectivity and node features rather than directly consuming the edge weights in the forward pass.

Repeating this procedure over time yields a sequence of dated graph-label pairs (G_t, y_t) . Each graph summarizes the recent cross-sectional structure of equity returns at date t , while the label measures market volatility over the future 30-trading-day horizon. The canonical graph-construction and model-configuration settings are reported in Appendix B.

3.3 Model Selection, Estimation, and Forecast Evaluation

3.3.1 GNN Architecture and Analytical Justification

The canonical model is a graph-level GraphSAGE regressor trained to predict the logarithm of forward 30-trading-day realized market volatility. Each supervised observation is one dated graph snapshot and one scalar market-level target. The model therefore maps (G_t, y_t) into a single forecast of future aggregate volatility, rather than forecasting volatility separately for each individual stock.

GraphSAGE is appropriate for this setting because the input data are relational. A standard tabular model would require the graph to be compressed into a fixed set of manually selected network statistics before estimation. By contrast, a GNN can process node-level information together with the local graph neighborhoods implied by the return-dependence structure. GraphSAGE is particularly suitable because it updates node representations by aggregating information from neighboring nodes, allowing the model to learn from both firm-level features and cross-sectional relationships. After message passing, global pooling converts the node embeddings into a market-level graph representation that can be mapped into the volatility forecast.

The canonical architecture uses four GraphSAGE message-passing layers with hidden dimension 256 and ReLU activations. Node embeddings are aggregated using global mean pooling. The pooled graph representation is passed through a multilayer perceptron consisting of a 256-unit hidden layer, ReLU activation, dropout with rate 0.1, and a final linear output layer. The output is a single scalar prediction for $y_t = \log(RV_{30,t}^{fwd})$.

This architecture is not treated as theoretically optimal in an absolute sense. Its role is to provide a scalable and interpretable graph-learning baseline that matches the structure of the research question. The aim is to test whether graph-based representations of the equity cross-

section contain predictive information for aggregate volatility, not to claim that GraphSAGE is the only possible architecture. Alternative configurations, including different graph-construction windows and architecture choices, are summarized in Appendix E.

3.3.2 Model Selection and Overfitting Control

Model selection is based on a chronological training-validation-test design. The training sample is used to estimate model parameters, the validation sample is used for model selection and early stopping, and the test sample is reserved for final out-of-sample evaluation. This design is essential because the model is flexible and could otherwise overfit historical volatility patterns.

The model is trained using mean squared error in log-target space, consistent with the transformed forecasting target. Parameters are estimated using the Adam optimizer with learning rate 5×10^{-6} , batch size 32, and a maximum of 100 epochs. The checkpoint with the lowest validation loss is retained as the final model. Early stopping is applied with a patience of 10 epochs, so training stops if validation performance fails to improve for ten consecutive epochs. No test-sample information is used in model selection. The canonical results are reported for the configured random seed used in the final pipeline rather than as averages across repeated random seeds. The reported performance should therefore be interpreted as the performance of the selected canonical specification, not as a distribution of outcomes across stochastic training runs.

Several additional choices are intended to control overfitting. Feature scaling is estimated on the training sample only and then applied unchanged to the validation and test samples. The validation period is chronologically later than the training period, making it a more realistic tuning environment than a random split. A placebo-graph ablation is also included to test whether performance depends strongly on the original graph topology or whether similar gains can be obtained after disrupting the network structure. Together, these design choices make the model-selection process more conservative and reduce the risk that the reported performance reflects in-sample fitting rather than genuine out-of-sample predictability.

3.3.3 Forecast Alignment and Evaluation Metrics

Forecast evaluation requires that the GNN forecast, realized-volatility target, VIX benchmark, HAR benchmark, and placebo forecast be compared on the same dated observations. The evaluation panel is therefore constructed by exact date alignment. Observations are retained only when all relevant forecasts and the realized target are available.

This ensures that differences in forecast loss are not driven by different sample compositions across models.

The GNN is estimated in log space, so log-scale errors are central to evaluation. The main log-space metrics are mean squared error and mean absolute error:

$$MSE_{log} = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2$$

$$MAE_{log} = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t|$$

The analysis also evaluates forecasts using variance-based QLIKE, a standard loss function in volatility-forecast evaluation. Since the empirical target and forecasts are expressed as annualized volatility levels, realized and forecast variance proxies are obtained by squaring the corresponding volatility quantities. Let v_t denote the realized forward volatility level and let $\hat{v}_t$ the corresponding forecast volatility level. The QLIKE loss is computed as:

$$QLIKE_t = \log(\hat{h}_t) + \frac{h_t}{\hat{h}_t}$$

where $h_t = v_t^2$ and $\hat{h}_t = \hat{v}_t^2$. Lower values indicate better forecast performance. This metric complements the log-space MSE and MAE measures by evaluating forecast accuracy in variance space, which is especially relevant in volatility-forecasting applications.

The benchmark set includes four forecast series. The GNN is the main graph-based model. The VIX is the primary forward-looking implied-volatility benchmark. The HAR-type model serves as a historical-volatility benchmark. In the implementation, this benchmark is a HAR-like log-realized-volatility comparator based on lagged realized-volatility components over short, medium, and longer horizons, with forecasts aligned to the same feature dates as the GNN and VIX forecasts. The placebo graph serves as an ablation designed to evaluate whether the trained model’s forecasts are sensitive to the original graph topology after the edge structure is perturbed while node features and labels are preserved. The full formal hypothesis tests focus primarily on the GNN-VIX comparison because the central research question asks whether equity-derived graph information is competitive with a strong market-implied benchmark. The HAR and placebo forecasts provide additional diagnostic context.

3.3.4 Robustness, Placebo Design, and Interpretability Diagnostics

The robustness design addresses whether the apparent forecasting value of the GNN is driven by specific market episodes or by the general structure of the model. This is important because volatility forecasts can appear successful if they perform unusually well during one

crisis period while adding less value in ordinary market conditions. To test this possibility, the thesis repeats the main GNN-VIX comparison on restricted test subsamples.

The canonical robustness samples are: the full test sample, the test sample excluding calendar year 2020, the test sample excluding the pre-specified stress window from 1 January 2022 to 31 October 2022, and the test sample excluding both 2020 and the stress window. These exclusions are applied only within the held-out test period. The purpose is not to remove inconvenient observations, but to examine whether the model’s relative performance survives when obvious extreme episodes are excluded. The formal mapping between hypotheses, test procedures, and restricted samples is summarized in Appendix C.

The placebo-graph ablation is a second robustness device. It keeps the dated node features and realized-volatility labels fixed while perturbing the original graph topology. In the canonical implementation, this perturbation is produced through a deterministic seeded permutation of edge endpoints, which disrupts the original correspondence between firms and their return-dependence neighborhoods. The trained graph model is then evaluated on these perturbed graph inputs. If the original graph performs much better than the placebo, that would support the claim that the baseline edge structure contributes meaningfully to forecast performance. If the placebo performs similarly, the interpretation must be more cautious: forecasting gains may arise from the broader cross-sectional representation, node-level information, or model architecture rather than from the exact baseline topology. This ablation is therefore useful for avoiding an overstatement of the network-specific contribution, but it should not be interpreted as a separately trained placebo model or as a complete test of all possible alternative graph structures.

Interpretability is addressed through forecast diagnostics rather than through a full causal explanation of the GNN’s internal weights. The thesis examines forecast variance, correlation with realized log-volatility, Mincer–Zarnowitz calibration statistics, and forecast error across volatility quantiles. These diagnostics help answer whether the GNN is genuinely informative, whether it is excessively smooth, whether it is well calibrated, and whether its performance differs across volatility regimes. Supplementary calibration and regime diagnostics are reported in Appendix D.

Taken together, the robustness and interpretability design serves three purposes. First, it tests whether the GNN’s relative performance is stable outside extreme periods. Second, it evaluates whether the exact graph topology appears to matter through the placebo comparison. Third, it clarifies the economic interpretation of the forecasts by identifying whether the model performs better in ordinary, moderately stressed, or high-volatility states.

3.4 Hypotheses and Empirical Testing

This subsection states the four empirical hypotheses tested in the thesis. All formal inference is conducted on the held-out test sample or on restricted subsamples of the test period. Since the forecasting target is constructed from overlapping forward 30-trading-day windows, regression residuals and loss differentials may exhibit serial dependence. The formal tests therefore use heteroskedasticity- and autocorrelation-consistent inference, with lag length tied to the forecast horizon.

Let y_t denote the logarithm of forward 30-trading-day realized market volatility, let $\hat{y}_t^{GNN}$ denote the GNN forecast, and let $\hat{y}_t^{VIX}$ denote the VIX-based benchmark forecast transformed into log space. All forecast-comparison tests are conducted on date-aligned observations.

3.4.1 Hypothesis 1: Predictive Content of the GNN Forecast

The first hypothesis examines whether the GNN forecast contains predictive information about future market volatility.

H_0 : the GNN forecast contains no predictive information about future realized volatility

H_1 : the GNN forecast contains predictive information about future realized volatility

This hypothesis is tested using a Mincer–Zarnowitz-type predictive regression on the held-out test sample:

$$y_t = \alpha + \beta \hat{y}_t^{GNN} + u_t$$

The null hypothesis is $H_0 : \beta = 0$. Rejection of this null indicates that the GNN forecast is systematically related to subsequently realized volatility. This test establishes whether the model is informative before considering whether it outperforms competing benchmarks.

3.4.2 Hypothesis 2: Relative Forecast Accuracy Against VIX

The second hypothesis examines whether the GNN achieves lower out-of-sample forecast loss than the VIX benchmark.

H_0 : the GNN does not achieve lower out-of-sample forecast loss than VIX

H_1 : the GNN achieves lower out-of-sample forecast loss than VIX

For each test date, a loss differential is constructed as:

$$d_t^{(L)} = L(y_t, \hat{y}_t^{VIX}) - L(y_t, \hat{y}_t^{GNN})$$

where $L(\cdot)$ is a forecast-loss function. Under this definition, positive values indicate lower loss for the GNN. The thesis evaluates this comparison using variance-based QLIKE and mean squared error in log space. The mean loss differential is tested using a Diebold–Mariano-type

comparison with HAC-adjusted variance estimates. Because the hypothesis is directional, the primary comparison is one-sided.

3.4.3 Hypothesis 3: Incremental Information Beyond VIX

The third hypothesis asks whether the GNN contains predictive information beyond that already embedded in the VIX.

H_0 : conditional on VIX, the GNN forecast adds no additional predictive information

H_1 : conditional on VIX, the GNN forecast adds additional predictive information

This hypothesis is tested using an augmented predictive regression:

$$y_t = \alpha + \beta_1 \hat{y}_t^{GNN} + \beta_2 \hat{y}_t^{VIX} + u_t$$

The relevant null hypothesis is $H_0 : \beta_1 = 0$. If the GNN coefficient remains statistically significant after controlling for VIX, then the graph-based forecast can be interpreted as containing incremental information beyond the implied-volatility benchmark. This test is more demanding than the relative-loss comparison because it asks whether the GNN contributes independent conditional information rather than simply achieving lower average forecast error.

3.4.4 Hypothesis 4: Robustness Outside Extreme Market Periods

The fourth hypothesis examines whether the GNN's apparent forecasting advantage is driven entirely by exceptional market episodes.

H_0 : Any apparent forecasting advantage of the GNN disappears once extreme periods are excluded

H_1 : the GNN retains its relative forecasting advantage after extreme periods are excluded

This hypothesis is evaluated by repeating the relative-accuracy and incremental-information tests on restricted test subsamples. The main exclusions remove calendar year 2020, the pre-specified 2022 stress window, and the union of both exclusions. If the GNN continues to perform favorably after these restrictions, the evidence suggests that its predictive usefulness is not purely a crisis-period artifact. If the advantage disappears, the interpretation becomes narrower and more episode-specific.

The four hypotheses together define the empirical logic of the thesis. Hypothesis 1 asks whether the GNN forecast is informative at all. Hypothesis 2 asks whether it improves forecast accuracy relative to VIX. Hypothesis 3 asks whether it adds information beyond VIX. Hypothesis 4 asks whether the results survive outside extreme periods. This sequence allows the thesis to move from basic predictability to benchmark comparison, incremental information, and robustness.

4 Analysis

4.1 Main Out-of-Sample Comparison

Table 4.1-A reports the main out-of-sample forecasting results for the held-out full test sample. The principal comparison is between the graph neural network (GNN) and the VIX benchmark, while two additional comparators are also shown: a HAR-type historical-volatility benchmark and a placebo-graph ablation in which the model architecture and node features are held fixed but the graph structure is intentionally disrupted. This broader comparison is useful because it allows the results to be interpreted from three angles at once: against a strong forward-looking market benchmark, against a standard historical-volatility model, and against a counterfactual designed to weaken the contribution of return-network structure.

model	mse_log	mae_log	qlike	n_samples
gnn	0.148179	0.28315	7.0893	1480
vix	0.199449	0.361865	6.908771	1480
har	0.156328	0.281466	7.128342	1480
placebo	0.147805	0.281627	7.082399	1480

Table 4.1-A: Out-of-sample forecast accuracy on the full test sample for the GNN, VIX, HAR, and placebo forecasts; lower values indicate better forecast performance. QLIKE is computed in variance space.

On the full test sample, the GNN performs clearly better than the VIX when forecast errors are evaluated in logarithmic space. The model’s mean squared error in logs is 0.1482, compared with 0.1994 for the VIX, while mean absolute error in logs is 0.2831 for the GNN and 0.3619 for the VIX. This indicates that the graph-based model produces materially more accurate forecasts once the target is viewed on the same log scale used in model estimation. Under variance-based QLIKE, however, the comparison favors the VIX. The GNN’s QLIKE loss on the full test sample is 7.0893, whereas the corresponding VIX value is 6.9088. The main full-sample message is therefore not that the GNN uniformly dominates the implied-volatility benchmark, but rather that it improves forecast accuracy relative to VIX on log-scale error metrics while not outperforming the VIX on variance-based QLIKE.

This difference in forecast behavior is also visible in Figure 4.1-A, which plots the realized target together with the GNN forecast and the VIX benchmark over the test period. Consistent with the loss metrics, the GNN appears smoother than the VIX across much of the sample, while the VIX reacts more sharply during the most turbulent episodes.

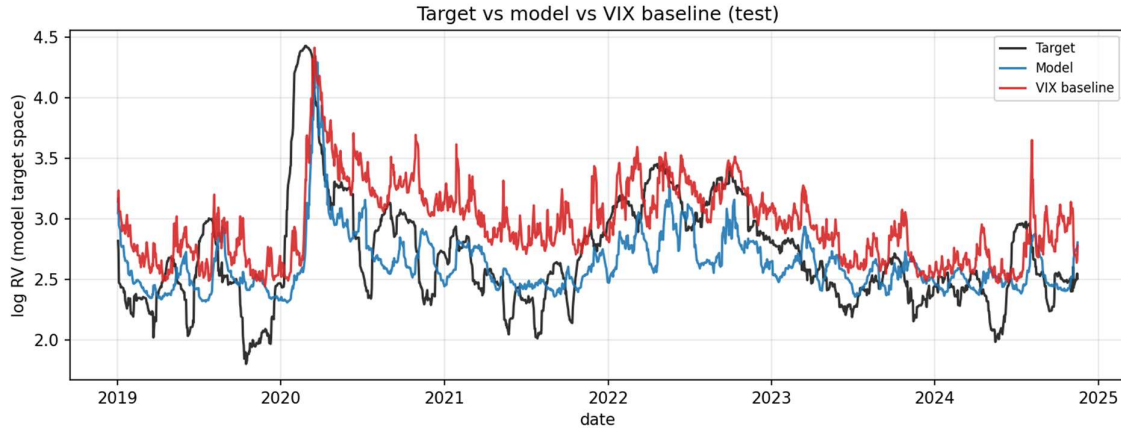


Figure 4.1-A: Test-period paths of realized log-volatility, the GNN forecast, and the VIX benchmark; the GNN is smoother across most of the sample, whereas the VIX reacts more sharply during turbulent episodes.

The comparison with the HAR benchmark yields a related conclusion. On the full test sample, the HAR model records an `mse_log` of 0.1563, which is worse than the GNN's 0.1482 but still materially better than the VIX. Under variance-based QLIKE, however, HAR performs worse than both the GNN and the VIX, with a value of 7.1283. This suggests that historical realized-volatility dynamics continue to contain useful predictive information, as the literature would lead one to expect, but that the GNN remains competitive even against a benchmark specifically designed to summarize persistence across volatility horizons. Put differently, the GNN does not owe its apparent success solely to comparison with an unusually difficult option-based benchmark; it also compares favorably with a more conventional academic baseline on log-scale forecast accuracy.

The placebo-graph comparison is more cautionary. On the full test sample, the placebo forecast is almost indistinguishable from the GNN and is in fact marginally better on the headline summary metrics reported in Table 4.1-A, with `mse_log` of 0.1478 versus 0.1482 for the GNN and QLIKE of 7.0824 versus 7.0893. These differences are economically small, but they matter for interpretation. They imply that the present implementation does not yet provide strong evidence that the exact graph structure itself is the dominant source of forecasting gains. Instead, part of the predictive power may derive from the broader modeling framework, including the date-level forecasting design, the node-level inputs, and the model's ability to summarize cross-sectional information even when the exact graph topology is perturbed. This does not negate the usefulness of the GNN specification, but it does counsel against making an overly strong claim about the incremental value of the current graph construction.

The restricted subsamples already hint at a more favorable picture for the GNN relative to VIX, although the strength of that picture depends on the loss function. When 2020 is excluded,

the GNN’s `mse_log` falls to 0.0900, compared with 0.1495 for the VIX. Its variance-based QLIKE is also slightly lower than the VIX’s, at 6.4515 versus 6.4575, although the formal QLIKE loss-differential test reported later does not provide statistically meaningful support for GNN superiority on that subsample. In the stress-stripped subsamples reported under Hypothesis 4, the GNN retains clear advantages over the VIX on log-scale forecast loss, and on the most restrictive union-exclusion sample it also achieves significantly lower QLIKE. This matters because it suggests that the model’s relative performance is not confined entirely to the most turbulent part of the COVID episode. At the same time, the placebo and HAR comparisons remain close enough that the results should be interpreted as evidence of a useful graph-based forecasting framework rather than definitive proof that the precise network topology extracted in the baseline specification is solely responsible for the gains.

Taken together, the main out-of-sample evidence supports three conclusions. First, the GNN contains meaningful predictive information and compares favorably with both VIX and HAR when forecast errors are measured in log space. Second, the model does not achieve blanket dominance over the VIX on all loss functions, since full-sample variance-based QLIKE favors the implied-volatility benchmark. Third, the placebo exercise shows that the current implementation’s gains should be interpreted with caution when attributing them specifically to network structure. For that reason, the remainder of this chapter examines the formal hypothesis tests and additional diagnostics in more detail, focusing not only on average forecast loss but also on predictive content, incremental information relative to VIX, robustness outside extreme periods, and the possibility that relative performance differences reflect calibration or smoothness rather than uniformly better tracking of volatility dynamics.

4.2 Hypothesis 1: Predictive Content of the GNN Forecast

Hypothesis 1 asks whether the GNN forecast contains predictive information about future 30-trading-day realized market volatility. In the methodology, this hypothesis was defined through a Mincer–Zarnowitz-type regression in which realized log-volatility is regressed on a constant and the GNN forecast in log space. Rejection of the null hypothesis that the slope coefficient equals zero indicates that the forecast is systematically related to subsequently realized volatility rather than behaving like an uninformative signal.

The evidence in favour of Hypothesis 1 is very strong. On the full held-out test sample, the estimated predictive regression is:

$$y_t = 0.1454 + 0.9719\hat{y}_t^{GNN} + u_t$$

with heteroskedasticity- and autocorrelation-consistent standard errors. The estimated slope coefficient on the GNN forecast is 0.9719, with a HAC standard error of 0.1104 and a t-statistic of 8.807. The corresponding p-value is approximately 1.29×10^{-18} , which is far below any conventional significance threshold. The null hypothesis that the GNN forecast contains no predictive information is therefore decisively rejected.

hypothesis_id	inference_procedure	loss_name	statistic	p_value_primary	tail
H1	mz_gnn		8.806689	1.29E-18	two_sided
H2	dm_loss_diff	qlike	-1.08459	0.860949	upper
H2	dm_loss_diff	mse_log	1.923563	0.027205	upper
H3	incremental_gnn_vix		1.94558	0.051705	two_sided

Table 4.2-A: Formal full-test results for Hypotheses 1–3; HAC-adjusted statistics and p-values are reported for predictive content, relative forecast accuracy, and incremental information beyond VIX.

This result is important because it establishes the most basic requirement for the thesis before any benchmark comparison is considered. A forecasting model does not need to dominate every comparator on every loss function in order to be informative, but it must at least produce forecasts that move meaningfully with the future variable it is intended to predict. Hypothesis 1 shows that this condition is satisfied very clearly. The GNN forecasts are strongly and systematically associated with subsequently realized market volatility in the pseudo-out-of-sample test period.

The coefficient estimates also help interpret the quality of that predictive relationship. The slope estimate is very close to one, suggesting that, on average, changes in the GNN forecast translate into proportionate changes in realized log-volatility rather than being heavily attenuated. The intercept estimate is 0.1454, with a HAC standard error of 0.2967 and a t-statistic of only 0.490, indicating that the constant term is not statistically distinguishable from zero. Although Hypothesis 1 is formally concerned with predictive content rather than full forecast unbiasedness, this pattern is encouraging because it suggests that the model’s forecasts are not obviously distorted by a large systematic level shift. The regression also yields an R^2 of 0.2915, implying that the GNN forecast explains a meaningful share of variation in the realized log-volatility target over the test sample.

This finding matters for the interpretation of the broader thesis. The mixed benchmark results reported in Section 4.1 do not imply that the GNN is weak or uninformative. Rather, they show that an informative model can still compare differently across alternative loss functions and across volatility regimes. Hypothesis 1 confirms that the GNN is a genuine predictive signal. The remaining issue is therefore not whether the model predicts future

volatility at all, but how strong and robust that predictive ability is relative to VIX, HAR, and the placebo-graph benchmark.

Taken together, the evidence strongly supports Hypothesis 1. The graph-based forecasting model contains clear out-of-sample predictive information about one-month-ahead realized market volatility. This provides the empirical foundation for the more demanding comparative tests that follow, where the analysis turns from predictive content in isolation to relative forecast accuracy, incremental information beyond the VIX, and robustness outside extreme market episodes.

4.3 Hypotheses 2 & 3: Comparison with VIX and Incremental Information

The corresponding full-test inferential results for Hypotheses 2 and 3 are reported in Table 4.2-A. Hypotheses 2 and 3 address the main comparative claims of the thesis. As set out in the methodology, Hypothesis 2 asks whether the GNN achieves lower out-of-sample forecast loss than the VIX benchmark, using a Diebold–Mariano-type comparison of mean loss differentials. The comparison is conducted using variance-based QLIKE and squared error in log-target space. Hypothesis 3 then asks whether the GNN contains incremental predictive information beyond that already embedded in the VIX, using an augmented predictive regression of realized log-volatility on both the GNN forecast and the VIX forecast in log form.

The full-test evidence for Hypothesis 2 is mixed but informative. On the one hand, the GNN performs clearly better than the VIX when forecast accuracy is assessed in log-target space. The full-test mse_log is 0.1482 for the GNN, compared with 0.1994 for the VIX, and the corresponding one-sided Diebold–Mariano-type test yields a HAC t-statistic of 1.924 with a p-value of 0.0272. This provides statistically significant evidence that the GNN delivers lower log-scale forecast loss than the implied-volatility benchmark. On the other hand, the variance-based QLIKE comparison does not support the same conclusion on the unrestricted test sample. The GNN’s full-test QLIKE is 7.0893, whereas the VIX records 6.9088, and the Diebold–Mariano-type test gives a HAC t-statistic of -1.085 with a one-sided p-value of 0.8609. Since the GNN is not better on variance-based QLIKE in the full sample, Hypothesis 2 is only partially supported. The appropriate interpretation is therefore narrower than blanket benchmark dominance: the GNN improves out-of-sample accuracy relative to VIX on log-scale forecast errors, but not on full-sample QLIKE.

Hypothesis 3 asks whether the GNN adds information beyond the VIX once both forecasts are considered jointly. On the full test sample, the estimated incremental regression is

$$y_t = 0.0134 + 0.2766\hat{y}_t^{GNN} + 0.6622\hat{y}_t^{VIX} + u_t$$

with HAC-adjusted standard errors. The coefficient on the GNN forecast is 0.2766, with a HAC standard error of 0.1422 and a t-statistic of 1.946, corresponding to a two-sided p-value of 0.0517. The coefficient on the VIX forecast is 0.6622, with a HAC standard error of 0.1316 and a t-statistic of 5.033, while the regression R^2 is 0.3671. These estimates indicate that both forecasts are positively related to future realized log-volatility in the joint specification, but the VIX remains the more strongly identified predictor in the full-sample conditional regression.

The implication for Hypothesis 3 is again qualified rather than absolute. The GNN coefficient is economically meaningful and very close to conventional statistical significance, but it does not cross the standard 5 percent threshold on the full test sample. As a result, the evidence for incremental information beyond VIX should be described as borderline rather than definitive in the baseline specification. This distinction matters for the overall thesis claim. The results do not support the stronger statement that the GNN unambiguously dominates the VIX or consistently renders it redundant. Instead, they suggest that the GNN contains useful information that may complement the VIX, while leaving some uncertainty about how robust that incremental contribution is in the full sample.

Taken together, Hypotheses 2 and 3 support a balanced interpretation of the main findings. The GNN clearly compares favorably with the VIX on log-scale forecast loss, which is consistent with the model’s training target and with the idea that graph-based cross-sectional signals contain genuine forecasting value. At the same time, the lack of full-sample QLIKE superiority and the borderline nature of the incremental-information test mean that the results should not be framed as universal benchmark dominance. The strongest defensible conclusion at this stage is that the GNN is competitive with, and in some dimensions superior to, the VIX benchmark, while the question of whether it adds robust conditional information beyond VIX remains more sensitive to sample choice. This motivates the next section, which examines whether the comparative evidence becomes stronger once extreme market episodes are excluded from the evaluation sample.

4.4 Hypothesis 4: Robustness Outside Extreme Market Periods

Hypothesis 4 asks whether any apparent forecasting advantage of the GNN is driven entirely by exceptional periods of market stress, or whether the model retains useful comparative performance once such periods are excluded. As specified in the methodology, this robustness exercise is implemented by repeating the relative-accuracy and incremental-information tests on restricted test subsamples, including a sample excluding calendar year 2020 and additional stress-window exclusions that overlap the held-out test period. The purpose

of this design is to distinguish general predictive usefulness from gains that might arise only during unusually turbulent episodes.

Table 4.4-A reports the H4 robustness results for the three restricted test samples. It summarizes the HAC-adjusted loss-comparison and incremental-information tests after excluding 2020, the pre-specified stress window, and the union of both periods.

The results show that the answer depends on which aspect of performance is being emphasized. When 2020 is excluded from the test sample, the GNN remains clearly superior to the VIX on log-scale forecast loss. Its mse_log falls to 0.0900, compared with 0.1495 for the VIX, and the corresponding one-sided Diebold–Mariano-type test yields a HAC t-statistic of 2.624 with a p-value of 0.00435. Under variance-based QLIKE, the GNN is only marginally lower in average loss than the VIX, with values of 6.4515 and 6.4575 respectively, and the formal loss-differential test provides no clear evidence of GNN superiority. The HAC t-statistic is 0.116, with a one-sided p-value of 0.4537. These findings suggest that the GNN’s comparative advantage after excluding 2020 is strong in log space but not in QLIKE.

sample	inference_procedure	loss_name	statistic	p_value_primary	n_obs
test_excl_2020	dm_loss_diff	qlike	0.116261	0.453723	1227
test_excl_2020	dm_loss_diff	mse_log	2.623608	0.00435	1227
test_excl_2020	incremental_gnn_vix		1.543363	0.122743	1227
test_excl_stress	dm_loss_diff	qlike	-0.66197	0.746006	1271
test_excl_stress	dm_loss_diff	mse_log	3.101275	0.000963	1271
test_excl_stress	incremental_gnn_vix		1.988418	0.046766	1271
test_excl_union	dm_loss_diff	qlike	3.611855	0.000152	1018
test_excl_union	dm_loss_diff	mse_log	5.464376	2.32E-08	1018
test_excl_union	incremental_gnn_vix		0.534614	0.592917	1018

Table 4.4-A: Robustness results for Hypothesis 4 under restricted test samples; HAC-adjusted statistics and p-values are reported after excluding 2020, the pre-specified stress window, and the union of both periods.

The incremental-information test on the test_excl_2020 sample is weaker. In the augmented regression including both the GNN forecast and the VIX forecast, the coefficient on the GNN forecast is 0.2735 with a HAC standard error of 0.1772, producing a t-statistic of 1.543 and a two-sided p-value of 0.1227. By contrast, the VIX coefficient remains large and precisely estimated at 0.7711 with a t-statistic of 5.266. This means that, after excluding 2020, the GNN continues to compare favourably with the VIX in terms of average log-scale forecast loss, but the evidence that it contributes additional conditional information beyond the VIX is not statistically strong on this subsample.

A second robustness exercise removes the pre-specified stress window that overlaps the test period. On this test_excl_stress sample, the evidence remains favorable to the GNN in log space. The Diebold–Mariano-type test for mse_log yields a HAC t-statistic of 3.101 with a one-

sided p-value of 0.00096, again showing strong improvement relative to VIX in log-target space. The QLIKE comparison is not favorable to the GNN on this subsample. The HAC t-statistic is -0.662, with a one-sided p-value of 0.7460, so there is no evidence that the GNN achieves lower QLIKE loss than VIX after excluding the stress window. More importantly, the incremental-information regression becomes statistically significant on this restricted sample. The coefficient on the GNN forecast rises to 0.3339, with a HAC standard error of 0.1679 and a t-statistic of 1.988, corresponding to a two-sided p-value of 0.0468. The VIX coefficient remains positive and highly significant at 0.5265. This is one of the stronger pieces of evidence in the robustness analysis, because it suggests that once the specific stress window is removed, the GNN still contributes incremental information beyond the benchmark even though the QLIKE loss comparison does not favor it.

The most restrictive robustness exercise removes both 2020 and the overlapping stress window, yielding the `test_excl_union` sample. On this subsample, the relative-loss results become especially strong. The Diebold–Mariano-type test rejects the null for both QLIKE and `mse_log`, with HAC t-statistics of 3.612 and 5.464, and one-sided p-values of approximately 1.52×10^{-4} and 2.32×10^{-8} , respectively. In other words, once the most extreme periods are jointly excluded, the GNN substantially outperforms the VIX on both variance-based QLIKE and the supplementary log-space criterion. However, the incremental-information regression tells a different story on this same sample. The coefficient on the GNN forecast falls to 0.0964, with a HAC standard error of 0.1803 and a t-statistic of only 0.535, while the VIX coefficient remains significant at 0.4703. Thus, on the union-exclusion sample, the GNN clearly improves average forecast loss relative to the benchmark, but this improvement does not translate into a statistically significant incremental-information result once both forecasts enter the regression together.

Taken together, the evidence supports Hypothesis 4 in an important but qualified sense. The GNN’s comparative performance is not purely a crisis-year artifact. In particular, the model continues to outperform the VIX on log-scale error metrics after excluding 2020, after excluding the overlapping stress window, and after excluding both sets of periods simultaneously. On the most restrictive union-exclusion sample, it also outperforms the VIX on variance-based QLIKE. At the same time, QLIKE does not support GNN superiority on the 2020-exclusion or stress-window-exclusion samples, and the incremental-information results remain less stable across subsamples: they are insignificant when 2020 alone is excluded, significant when the overlapping stress window is excluded, and insignificant again on the union exclusion. The most defensible interpretation is therefore that the GNN’s relative

accuracy is reasonably robust outside extreme periods, especially in log-space loss, but the claim that it adds independent conditional information beyond the VIX remains more sample-sensitive.

This robustness analysis strengthens the overall thesis because it rules out the simplest version of a crisis-only explanation. The GNN's strongest gains are not confined entirely to the most obvious high-volatility episode in the sample. Nevertheless, the mixed pattern across loss functions and conditional regressions also reinforces the need for caution. The model appears to be genuinely useful outside extreme periods, but its advantage is clearer in average forecast-loss comparisons than in the more demanding test of whether it consistently adds information beyond a strong implied-volatility benchmark. That issue motivates the next subsection, which examines forecast calibration, smoothness, and performance across different volatility regimes in more detail.

4.5 Forecast Calibration and Variance Diagnostics

The preceding hypothesis tests establish that the GNN contains predictive information and, in several settings, compares favorably with the VIX benchmark. However, average forecast-loss comparisons alone do not fully explain why one forecast performs better than another. A model may appear successful either because it genuinely tracks the behavior of realized volatility more effectively or because it produces smoother forecasts that reduce average error in some regimes while failing to capture the largest volatility spikes. For this reason, the analysis also examines forecast variance, correlation with the realized target, Mincer–Zarnowitz calibration statistics, and forecast performance across target-volatility quantiles.

The first diagnostic concerns forecast variance in log space. On the full test sample, the GNN has substantially lower forecast variance than both HAR and VIX. The variance of the GNN forecast is 0.0623, compared with 0.1035 for VIX and 0.1022 for HAR. The placebo graph lies in between, at 0.0667. This pattern indicates that the GNN produces smoother forecasts than the market-implied and historical-volatility benchmarks. The same result appears when 2020 is excluded. In the `excl_2020` sample, the GNN forecast variance is 0.0316, compared with 0.0669 for VIX and 0.0657 for HAR. Thus, the graph-based model is systematically less volatile in its predictions than the two benchmark forecasts. This regime dependence is summarized in Figure 4.5-A, which plots log-scale forecast error across target-volatility quintiles on the full test sample.

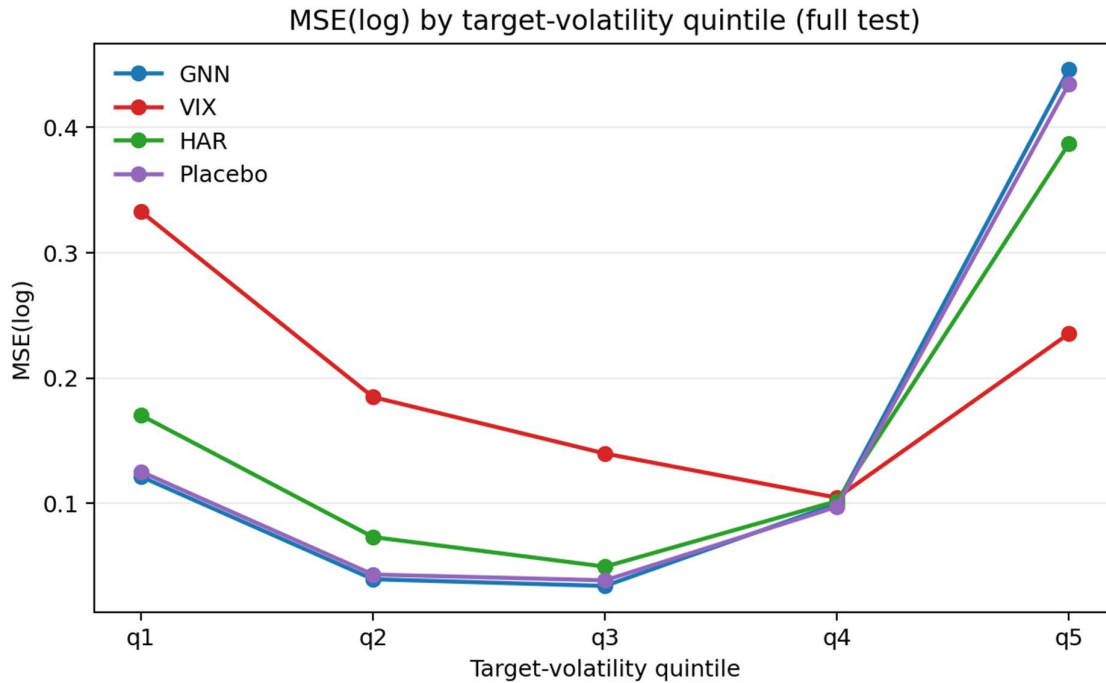


Figure 4.5-A: Log-scale forecast error by target-volatility quintile on the full test sample; the GNN and placebo perform better in lower and middle volatility regimes, while the VIX performs better in the highest-volatility quintile.

Smoother forecasts are not automatically better forecasts, so the next question is whether the lower variance reflects useful regularization or excessive compression. The correlation diagnostics show that the answer is mixed. On the full test sample, the GNN's correlation with realized log-volatility is 0.5399. This is meaningful and clearly indicates predictive relevance, but it remains below the corresponding VIX correlation of 0.5993. HAR also has a lower correlation than VIX at 0.5151, while the placebo graph is almost identical to the GNN at 0.5369. When 2020 is excluded, the same ordering remains. The GNN and placebo both have correlations close to 0.554, HAR rises slightly to 0.5685, and VIX remains the most strongly correlated with the target at 0.6590. These results suggest that VIX tracks the timing of realized volatility movements more closely than the GNN, even though it does not always deliver lower average forecast loss.

The Mincer–Zarnowitz calibration diagnostics help reconcile these findings. On the full test sample, the GNN has an intercept of 0.1454 and a slope of 0.9719, which is very close to one. By contrast, VIX has an intercept of 0.2233 and a slope of 0.8371, while HAR has an intercept of 0.7362 and a slope of only 0.7239. The placebo graph records an intercept of 0.2365 and a slope of 0.9339. These estimates indicate that the GNN forecasts are, on average, more proportionate to realized volatility than the VIX or HAR forecasts. In other words, although VIX is more tightly correlated with the target, its forecast changes appear somewhat less well

calibrated in magnitude, whereas the GNN's forecast movements are closer to one-for-one with realized outcomes. The same broad pattern holds in the `excl_2020` sample. There, the GNN slope rises to 1.1159, the placebo slope is 1.0501, the VIX slope is 0.9122, and the HAR slope is 0.7940. Again, the GNN does not appear to be a severely distorted or mechanically compressed predictor. Rather, it looks like a relatively well-calibrated but smoother forecast.

The most informative diagnostic is the performance breakdown by target-volatility quantile. On the full test sample, the GNN outperforms the VIX on `mse_log` in the lower four volatility quintiles. In quantile 1, the GNN's loss is 0.1213 compared with 0.3328 for VIX; in quantile 2, 0.0393 versus 0.1848; in quantile 3, 0.0340 versus 0.1397; and in quantile 4, 0.0997 versus 0.1044. However, the pattern reverses sharply in the highest-volatility quintile. In quantile 5, the GNN's `mse_log` rises to 0.4466, whereas the VIX records 0.2355. This implies that the GNN's full-sample advantage is not uniform across market states. The model is distinctly better in calm and moderately volatile periods, but it is weaker when volatility reaches its most extreme realized levels.

The same conclusion emerges even more clearly after excluding 2020. In the `excl_2020` sample, the GNN again dominates the VIX in the first four volatility quintiles, with especially large improvements in the lower and middle parts of the distribution. Yet in the highest-volatility quintile, the VIX becomes decisively better: the GNN's `mse_log` is 0.2158, whereas the VIX records only 0.0544. This shows that the model's relative advantage is not simply driven by the COVID period. Even outside 2020, the GNN remains stronger in ordinary and moderately stressed environments, while VIX retains a pronounced advantage in the most extreme volatility regime.

These diagnostics therefore lead to a more precise interpretation of the earlier hypothesis tests. The GNN does not appear to outperform VIX merely because it is an uninformative low-variance forecast. Its correlation with the target is substantial, and its Mincer–Zarnowitz slope is close to one, which is consistent with meaningful predictive content and reasonably good calibration. At the same time, the model's lower forecast variance is economically relevant. It seems to help reduce average error in low- and medium-volatility environments, but it also coincides with weaker performance when volatility enters the upper tail. In that sense, the GNN is better understood as a well-calibrated, smoother forecasting model that performs especially well outside the most extreme volatility states, rather than as a uniformly superior replacement for the VIX.

A final point concerns the placebo and HAR comparators. The placebo graph behaves very similarly to the GNN across all four diagnostics: its forecast variance, correlation, calibration

slope, and quantile-level errors are all close to those of the main model. HAR, by contrast, is more volatile and less well calibrated on average, but it performs relatively strongly in the highest-volatility quintile, especially once 2020 is excluded. This reinforces two broader conclusions. First, the current implementation’s gains cannot yet be attributed solely to the exact graph topology, since the placebo benchmark remains very competitive. Second, the strongest contribution of the GNN lies in producing stable and well-calibrated forecasts across most of the volatility distribution, even if the VIX remains more informative in the tail.

Taken together, the diagnostics strengthen the interpretation of the main results. They show that the GNN’s favorable log-scale performance is not just a trivial artifact of noise suppression, but neither does it imply universal dominance across all environments. The model appears most useful as a forecasting device for typical and moderately stressed market conditions, while the VIX continues to provide particularly valuable information in episodes of extreme realized volatility. This nuanced pattern is important for the discussion that follows, because it clarifies both the strengths and the limitations of graph-based volatility forecasting in the current empirical design.

5 Discussion

5.1 Interpreting the Main Findings

This thesis set out to examine whether graph-based representations of cross-sectional equity-return information can be used to improve forecasts of forward 30-trading-day realized market volatility. The results suggest that the answer is yes, but only in a qualified sense. The graph-based model clearly contains predictive information about future realized market volatility, compares favorably with the VIX on log-scale forecast errors, and retains a meaningful relative advantage in several robustness exercises. At the same time, the evidence does not support a stronger claim of universal benchmark dominance. In particular, the full-sample QLIKE comparison favors the VIX, the incremental-information result relative to VIX is borderline in the main test sample, and the VIX remains more informative in the highest-volatility regime. The strongest conclusion, therefore, is that the graph-based specification is a useful forecasting device, rather than that the exact baseline graph topology has been shown to be the dominant source of the model’s gains.

The first important implication is that the graph-based specification is genuinely informative rather than mechanically fitted. This is most clearly established by the predictive-content test, where the GNN forecast is strongly associated with subsequently realized log-volatility. That matters because it confirms that the model is not merely producing smooth market-level series that happen to look plausible *ex post*. Instead, the model output is systematically related to future market volatility in a held-out test period. This is a meaningful result in its own right, especially because the forecasting target is genuinely forward-looking and the empirical design enforces a clean no-look-ahead structure.

The second implication is that the strongest comparative evidence in favor of the GNN appears in log space rather than in the full-sample level-based loss comparison. This distinction is important for interpretation. Since the model is trained on the logarithm of forward realized volatility, it is not surprising that its clearest gains emerge when performance is evaluated on the same transformed scale. The log transformation stabilizes the target, reduces the influence of extreme observations, and makes proportional forecast errors more central than absolute deviations at the most turbulent points in the sample. In that sense, the GNN seems particularly well suited to forecasting the normal and moderately stressed evolution of volatility rather than the most extreme spikes.

This in turn helps explain why the GNN can outperform the VIX on average log-scale errors while failing to dominate it on full-sample QLIKE. QLIKE evaluates forecast accuracy

in variance space and is therefore more sensitive to large proportional errors in high-volatility states than the log-space MSE comparison. The fact that the VIX remains superior under this criterion suggests that the implied-volatility benchmark continues to be particularly valuable when markets move into the upper tail of the volatility distribution. Thus, the results do not imply that network-based forecasting makes option-implied information obsolete. Rather, they suggest that cross-sectional equity-market structure provides a useful and partially distinct forecasting signal whose strengths are clearest outside the most extreme volatility states.

5.2 What the Results Say about Network Structure

A central motivation of the thesis was the idea that aggregate indicators such as the VIX summarize expected market volatility without revealing how fragility is distributed across the equity market itself. The network approach was designed to address that limitation by representing the market as a cross-section of linked firms whose dependence structure evolves over time. Conceptually, this remains an attractive framing. If market co-movement becomes tighter, more concentrated, or more synchronized before aggregate volatility fully materializes, then graph-based models should in principle be able to convert that structural information into useful forecasts.

The empirical results provide partial support for that idea, but also impose an important qualification. The placebo-graph exercise shows that the current implementation does not yet isolate a large marginal contribution from the exact graph topology itself. Because the placebo benchmark remains very close to the main GNN across several performance measures, the present results do not justify a strong claim that the specific baseline graph construction is the dominant source of predictive gains. The safest interpretation is more modest: the graph-based framework appears useful, but the evidence does not yet show that the particular edge structure extracted by the baseline procedure is uniquely responsible for the model’s forecasting advantage.

This is not a trivial point. It changes the level at which the contribution should be understood. The thesis can credibly claim that graph-based modeling of the equity cross-section yields informative out-of-sample volatility forecasts and compares favorably with important benchmarks in several settings. It cannot yet claim, at least on the basis of the current placebo results, that the exact network structure is decisively more informative than closely related cross-sectional alternatives. In other words, the thesis provides stronger evidence for the usefulness of a graph-based forecasting design than for the indispensability of the exact baseline topology.

Even so, the placebo result should not be read as a failure. The fact that the placebo model is competitive may simply indicate that much of the useful signal is already contained in the cross-sectional organization of node-level information and in the broader graph-learning architecture, rather than in fine details of edge placement. This interpretation is still compatible with the broader thesis motivation. The market may contain forward-looking information in its cross-section without requiring the exact observed edge pattern to dominate every perturbed alternative. What the current results suggest is that the implementation captures useful market-wide structure, but that richer node features, stronger graph constructions, or architectures that exploit edge information more fully may be necessary to demonstrate a sharper network-specific effect.

5.3 Why VIX Still Matters

One of the clearest lessons from the analysis is that the VIX remains a demanding and meaningful benchmark. This is consistent with the literature review, which emphasized that implied-volatility measures often outperform purely historical models at short horizons. In the present thesis, the VIX retains two important strengths. First, it is more strongly correlated with realized log-volatility than the GNN. Second, it is much more effective in the highest-volatility regime, where the GNN’s errors rise sharply while the VIX remains comparatively accurate. This pattern is especially visible in the quantile-based diagnostics and persists even after excluding 2020.

Economically, this makes sense. The VIX is derived from option prices and therefore reflects information from market participants who are directly pricing protection against future tail risk. When uncertainty rises abruptly or market conditions become extreme, option markets may adjust faster and more aggressively than signals extracted solely from historical equity-return structure. The GNN, by contrast, is built from backward-looking return and dependence information, even if it learns those patterns in a flexible way. This makes it plausible that the model can provide cleaner and better-calibrated forecasts in ordinary states while still underreacting relative to the VIX when volatility enters the tail.

This interpretation also helps resolve an apparent tension in the results. At first glance, it may seem contradictory that the GNN is often better calibrated and performs better on log-scale forecast errors, while the VIX remains more correlated with the target and stronger in the upper tail. But these findings are compatible. The GNN appears to provide smoother and more proportionate forecasts across most of the sample, which improves average accuracy in lower and medium volatility states. The VIX, by contrast, appears to track the timing and scale of

extreme volatility shocks more aggressively. Thus, each forecast is strong in a different way. The GNN seems better suited to stable forecasting across typical market conditions, whereas the VIX remains particularly valuable as a barometer of severe turbulence.

This is an important interpretive outcome for the thesis. It means the contribution should not be framed as replacing the VIX, but rather as showing that cross-sectional equity-market structure can generate a useful complementary forecasting signal. That is a more defensible and more interesting result than an overstated claim of blanket superiority. In practical terms, it suggests that network-based models may be most valuable when the objective is to improve forecast stability and average accuracy over a broad range of conditions, while implied-volatility benchmarks retain special importance in periods of extreme stress.

5.4 Robustness and What It Really Means

The robustness analysis adds an important layer to this interpretation. Once extreme periods are removed, the GNN's relative performance often becomes stronger rather than weaker, especially in log-space loss. The model continues to outperform the VIX on log-scale Diebold–Mariano-type tests after excluding 2020, after excluding the overlapping stress window, and after excluding both sets of periods simultaneously. On the union-exclusion sample, the GNN also outperforms the VIX on both QLIKE and log-MSE. This is an important result because it rejects the simplest criticism that the model only looks good because of one unusual crisis episode. At the same time, QLIKE does not support GNN superiority on the 2020-exclusion or stress-window-exclusion samples individually, so the robustness claim remains stronger for log-space loss than for QLIKE across all restricted samples.

At the same time, the robustness results do not eliminate all ambiguity. The incremental-information tests remain sample-sensitive. The GNN coefficient becomes significant on the stress-window exclusion but not on the 2020 exclusion or the union exclusion. This suggests that the model's relative accuracy is more stable than its independently identified contribution conditional on VIX. Put differently, the GNN often improves forecast loss relative to the benchmark, but it does not always emerge as a clearly distinct source of incremental information once both forecasts enter the same regression. That is not a contradiction; it simply means that comparative predictive success and conditional incremental identification are not the same empirical claim.

This distinction is worth emphasizing because it sharpens the substantive contribution of the thesis. The most robust result is that the graph-based model is competitive with, and often superior to, the VIX in average forecast-loss terms outside extreme periods, especially in log

space. The weaker and more tentative result is that the model consistently adds independent information beyond what the VIX already contains. This is precisely why the thesis should use careful language such as “provides evidence” rather than stronger verbs such as “demonstrates” when summarizing its central findings.

5.5 Practical Implications for Risk Monitoring

The practical implication of the results is not that a graph-based forecast should replace established volatility indicators such as the VIX. A more defensible interpretation is that graph-based forecasts can provide an additional risk-monitoring signal alongside implied-volatility and historical-volatility benchmarks. This distinction matters because the empirical results show different strengths across market states. The GNN performs especially well in low- and medium-volatility environments and often delivers lower log-scale forecast errors than the VIX, while the VIX remains more informative in the highest-volatility regime. A practical user should therefore treat the graph-based forecast as complementary rather than substitutive.

In a risk-management setting, the graph-based forecast would be most useful as part of a monitoring dashboard that combines several types of information. The VIX would continue to summarize option-market expectations and tail-risk demand, the HAR-type forecast would summarize historical volatility persistence, and the graph-based model would summarize information contained in the cross-sectional structure of equity returns. The value of the graph-based signal is that it is derived from the organization of individual stocks rather than from index options alone. It may therefore help identify periods in which diversification is weakening or market co-movement is becoming more concentrated, even when aggregate indicators have not yet fully adjusted.

The disagreement between the graph-based forecast and the VIX may also be informative. If the VIX rises sharply while the graph-based forecast remains moderate, the signal may reflect option-market stress, hedging demand, or tail-risk pricing that is not yet visible in realized equity-return structure. If the graph-based forecast rises while the VIX remains comparatively contained, the signal may suggest that dependence within the equity cross-section is becoming more fragile even before option-implied volatility fully reflects that risk. In either case, the point is not that one signal is always correct, but that their divergence can help risk managers distinguish between different sources of market uncertainty.

These practical implications should be interpreted with model-governance caution. The placebo comparison shows that the current implementation does not isolate a decisive marginal contribution from the exact baseline topology, so the model should not be used as a precise map

of causal contagion or stock-level shock transmission. It should instead be treated as a forecasting tool whose outputs require ongoing benchmark comparison, recalibration, and stress-period monitoring. In operational use, its forecasts would need to be checked against implied-volatility indicators, historical-volatility models, and realized market outcomes over time. This is especially important because the model is less effective in the highest-volatility regime, where risk managers may place the greatest value on rapid warning signals.

The main managerial value of the thesis is therefore to show that equity-market structure can be transformed into a usable forecasting signal, but also to clarify the conditions under which that signal is most useful. The graph-based model appears most relevant for improving forecast stability and monitoring ordinary to moderately stressed market environments. For extreme volatility states, implied-volatility benchmarks remain essential. A practical risk-monitoring framework would therefore combine the graph-based forecast with VIX-based and historical-volatility signals rather than relying on any single model in isolation.

5.6 Limitations of the Current Empirical Design

Several limitations should be acknowledged. First, the baseline node-feature set is intentionally parsimonious. Using only rolling mean return and rolling volatility keeps the model interpretable and computationally manageable, but it also limits the amount of firm-level information available to the network. Richer node attributes, such as size, turnover, sector exposure, or alternative realized-risk measures, could allow the graph model to exploit cross-sectional structure more fully.

Second, the current graph construction uses a relatively simple dependence representation and sparsification rule. Although this is reasonable for a baseline specification, the predictive content of a financial network can depend materially on how links are measured and filtered. Different rolling windows, dependence measures, or graph representations could alter the strength of the network signal. This is especially relevant in light of the placebo comparison, which suggests that the exact baseline topology may not yet be the decisive source of forecasting gains.

Third, the architecture itself imposes limitations. The model produces one graph-level forecast per date and therefore captures the market as a sequence of dated snapshots rather than as a fully dynamic temporal graph. This is a sensible design for a bachelor's thesis and aligns with the forecasting target, but it may still leave time-dependent structure underexploited. In addition, the baseline implementation does not yet convert the proximity of firms in the graph into a richer economic interpretation beyond message passing and pooling.

Fourth, the benchmark comparison is deliberately demanding. The VIX is not a pure statistical expectation of realized volatility, since it reflects risk premia and option-market conditions, but it remains a very strong market-based predictor. Comparing an equity-return-based graph model against it is therefore informative, but it also sets a high standard that may partly explain why the incremental-information results are less stable than the relative-loss results.

These limitations do not undermine the thesis. Instead, they help locate its contribution more precisely. The thesis should be understood as showing that graph-based cross-sectional equity-return information is a useful forecasting input for forward realized market volatility, while also showing that the strongest claims about exact network topology and robust incremental dominance over VIX require further evidence.

5.7 Implications and Directions for Further Work

The results point toward several natural extensions. A first avenue is to strengthen the network-specific component of the model. This could involve richer node features, alternative dependence measures, stronger graph ablations, or architectures that make more explicit use of edge information. Such extensions would help clarify whether the forecasting gains come primarily from the general cross-sectional representation or from the specific network topology extracted from equity returns.

A direct continuation of this thesis is therefore to use the modular empirical codebase as the basis for a broader specification-search exercise. The accompanying GitHub repository was developed so that individual components of the pipeline, including graph construction, node-feature design, model architecture, benchmark evaluation, placebo testing, and robustness analysis, can be modified without rebuilding the full framework from the beginning (Kokiauri, 2026). I intend to continue extending this framework by running a substantially wider set of configurations, including alternative graph topologies, richer firm-level features, longer hyperparameter searches, repeated random-seed experiments, and more computationally demanding graph-learning architectures. If these future experiments identify a specification that materially improves forecast accuracy and isolates a stronger network-specific contribution, that extension could form the basis of a separate follow-up thesis or research paper.

A second avenue is benchmark expansion. The addition of a HAR-type benchmark already improves the thesis substantially by showing that the GNN compares favorably not only with an implied-volatility benchmark but also with a standard historical-volatility model. Future work could extend this by including additional econometric and machine-learning baselines,

which would help map more precisely the range of environments in which graph-based forecasting is especially useful.

A third avenue concerns forecast combination. The empirical results suggest that the GNN and the VIX are strong in different regimes rather than being perfect substitutes. This raises the possibility that combining the two signals may produce better overall forecasts than either model alone. Such an extension would be especially interesting because the current results already hint at complementarity: the GNN appears stronger in ordinary and moderately stressed conditions, whereas VIX remains especially informative in the upper tail.

Taken together, the discussion suggests that the thesis makes a credible and useful contribution, both empirically and practically, but one that is narrower and more nuanced than a claim of universal benchmark dominance. The graph-based model is informative, well calibrated, and often competitive with strong benchmarks. Its advantages are clearest on log-scale errors and outside the most extreme volatility states. Its limitations are equally informative, since they reveal where implied-volatility benchmarks continue to dominate and where the current implementation does not yet isolate a decisive network-topology effect. Framed in this way, the thesis contributes both a positive empirical result and a clearer understanding of what graph-based volatility forecasting can and cannot yet achieve.

6 Conclusion

This thesis examined whether graph-based representations of the cross-sectional dependence structure of equity returns can be used to forecast forward 30-trading-day realized market volatility. The motivation was that standard forecasting approaches, while often useful, capture different dimensions of market uncertainty. Historical volatility models summarize persistence in past realized variation, and implied-volatility measures such as the VIX provide a forward-looking benchmark derived from option prices. However, neither type of approach directly reveals how dependence is organized across firms within the equity market. If changes in that cross-sectional organization contain forward-looking information about fragility, then a graph-based forecasting model may be able to transform that information into useful predictions of subsequent aggregate volatility.

To study this question, the thesis constructed a sequence of dated graph snapshots for a stable U.S. equity universe and used those graphs to train a graph neural network that produces one market-level volatility forecast per date. The forecasting target was forward 30-trading-day realized market volatility, modeled in logarithmic form and evaluated in a strictly chronological out-of-sample design. The resulting forecasts were compared with the VIX as the main forward-looking benchmark, with additional reference to a HAR-type historical-volatility model and a placebo-graph ablation. Formal inference was then conducted through predictive-content, relative-accuracy, incremental-information, and restricted-sample robustness tests, complemented by calibration and volatility-regime diagnostics.

The results show that the graph-based model contains genuine predictive information about future realized market volatility. This is the clearest and strongest finding of the thesis. The GNN forecast is strongly and systematically related to subsequently realized log-volatility in the held-out test sample, which means that the model is not merely producing smooth or plausible-looking series without predictive relevance. More demanding benchmark comparisons also provide meaningful support for the graph-based approach, although in a qualified way. In particular, the GNN outperforms the VIX on log-scale forecast errors in the main test sample and in several restricted-sample robustness exercises. This indicates that cross-sectional equity-return information can be converted into a useful out-of-sample forecasting signal within a graph-based forecasting framework.

At the same time, the thesis does not support a stronger claim of universal benchmark dominance. The full-sample QLIKE comparison favors the VIX rather than the GNN, and the incremental-information regression beyond VIX is borderline in the unrestricted test sample

and unstable across restricted subsamples. In addition, the calibration and quantile diagnostics show that the GNN performs especially well in low- and medium-volatility environments, while the VIX remains clearly superior in the highest-volatility regime. These results imply that the graph-based model should not be interpreted as a wholesale replacement for implied-volatility benchmarks. Rather, it should be understood as a complementary forecasting device whose strengths differ from those of the VIX.

The placebo-graph comparison adds an additional qualification. Because the placebo benchmark remains close to the main GNN across several metrics, the current implementation does not yet isolate a large and decisive marginal contribution from the exact baseline graph topology. This means that the thesis can more confidently claim that graph-based modeling of cross-sectional equity-return information is useful for forecasting than that the precise network structure extracted in the baseline specification is uniquely responsible for the gains. That distinction is important, because it locates the contribution of the thesis at the level of a promising and informative graph-based forecasting framework rather than at the level of a fully settled structural mechanism.

Even with these qualifications, the thesis makes a meaningful contribution. Substantively, it provides evidence that graph-based representations of cross-sectional equity-return information contain useful out-of-sample forecasting value for future aggregate volatility beyond what is captured by purely historical univariate models. Methodologically, it shows how graph construction, graph neural networks, strong benchmark comparison, placebo ablation, formal hypothesis testing, and calibration diagnostics can be combined in a coherent empirical framework. The results therefore suggest that graph-based approaches built from cross-sectional equity-return information can contribute to forecasting forward market volatility, while leaving open how much of that value depends on exact baseline topology.

Several limitations remain. The node-feature set is deliberately narrow, the baseline graph construction is relatively simple, and the canonical GraphSAGE implementation does not directly exploit edge weights in the forward pass. The model also operates on dated graph snapshots rather than on a fully dynamic temporal graph representation. These limitations likely matter for the strength of the estimated network effect and help explain why the placebo benchmark remains competitive. They also define the next stage of the research agenda. The accompanying GitHub codebase makes it possible to continue the project beyond the canonical specification reported in this thesis by testing a wider range of graph designs, node features, model architectures, tuning procedures, benchmark combinations, and forecast-combination methods that explicitly integrate graph-based and implied-volatility signals (Kokiauri, 2026).

My intention is to continue running additional configurations after this thesis, especially specifications that require greater computational capacity than was available during the present project. If a future configuration produces substantially stronger and more robust evidence of network-specific predictive value, that work could justify a separate follow-up study focused specifically on optimizing graph-based volatility forecasting.

Overall, the evidence supports a careful but positive conclusion. Graph-based representations of cross-sectional equity-return structure can be used to generate informative out-of-sample forecasts of forward realized market volatility. Their strongest gains appear on log-scale error metrics and outside the most extreme volatility states, while the VIX remains particularly valuable in tail-risk environments. The thesis therefore does not show that network-based forecasting dominates traditional benchmarks in every sense. What it does show is that cross-sectional equity-return information can be organized and modeled in a graph-based framework that yields useful forward-looking forecasting signals, even if the current evidence does not yet show that the exact baseline graph topology is uniquely responsible for that predictive value.

7 Appendix

7.1 Appendix A – Data, Sample, and Universe

input_file	source	role_in_pipeline	frequency	notes
WRDS.csv	CRSP daily security-level data via WRDS (or equivalent vendor extract).	Equity panel input: daily stock returns and identifiers for the cross-section, universe construction, rolling window features, and correlation-based graph edges.	Daily (U.S. equity trading days).	Security-by-date panel with CRSP-style identifiers and return fields; not used for the VIX benchmark.
snp500_volatility.csv	Public index series for the S&P 500 cash index (vendor of record as obtained by the researcher).	Target construction input: daily index returns for forward-looking realized volatility on the aggregate market.	Daily.	Calendar dates must align with the evaluation spine so targets and benchmarks are comparable.
VIX.csv	CBOE VIX (e.g., FRED or CBOE release), or equivalent.	Benchmark input only: observed market-implied volatility joined for out-of-sample comparison; excluded from model features by design.	Daily.	Used in evaluation and reporting, not as a predictor in the graph model.

Table A.1. Data inputs and their roles in the canonical thesis pipeline

field_name	required	description	used_for
date	Yes	Trading calendar date for each security observation.	Panel identity; time alignment across inputs.
permno	Yes	CRSP permanent security identifier.	Panel identity; cross-sectional linking.
permco	Yes	CRSP permanent company identifier.	Firm-level identifier consistency.
shrcd	Yes	CRSP share code classifying the security type.	Universe maintenance (ordinary common shares).
exchcd	Yes	CRSP exchange code for the listing venue.	Universe maintenance and listing filters.
ret	Yes	Daily total return including distributions.	Return construction; dividend-inclusive comparison series.
retx	Yes	Daily return excluding distributions.	Return construction; principal return series for rolling features and dependence in the canonical pipeline.
prc	Yes	CRSP price field (closing or bid-ask composite per vendor convention).	Market-capitalization ranking with shares outstanding.
vol	Yes	Daily trading volume.	Auxiliary trading-activity information in the equity panel.
shrout	Yes	Shares outstanding (CRSP units).	Market-capitalization ranking for universe selection.
dlstcd	Yes	CRSP delisting code.	Delisting and survivorship coding.

dlret	Yes	Delisting-period return.	Retained raw delisting field; not incorporated into canonical return construction.
dlretx	Yes	Delisting return excluding distributions.	Retained raw delisting field; not incorporated into canonical return construction.
dlprc	Yes	Delisting price.	Retained delisting metadata; not used in the canonical return construction.
dlpdt	Yes	Last quoted price date before delisting.	Retained delisting-date metadata; not used in the canonical return construction.
ticker	Yes	Listed ticker symbol.	Security identification (auxiliary).
comnam	Yes	Company name.	Security identification (auxiliary).
cusip	Yes	CUSIP identifier.	Security identification (auxiliary).
ncusip	Yes	Numeric CUSIP.	Security identification (auxiliary).
secstat	Yes	Security trading status or header flag from the vendor file.	Security metadata retained from the extract.

Table A.2. WRDS/CRSP raw-input schema used in the canonical empirical pipeline

input_file	field_name	required	description	used_for
snp500_volatility.csv	date	Yes	Trading or observation date for the index series (may appear under standard vendor date headers).	Target construction; calendar merge to the modeling spine.
snp500_volatility.csv	sprtrn	Yes	Daily S&P 500 index total return.	Target construction; forward realized variance of the index.
VIX.csv	date	Yes	Observation date for the VIX quote (may appear under standard vendor date headers).	Benchmark alignment; calendar merge to forecasts and targets.
VIX.csv	vix	Yes	CBOE VIX closing level (vendors may label the series VIXCLS).	Benchmark input; out-of-sample comparison only.

Table A.3. Required target and benchmark raw-input schema for reproducing the empirical pipeline

7.2 Appendix B – Targets, Benchmarks, and Model Setup

component	setting	canonical_value
Graph construction	Historical window length	20 trading days

Graph construction	Minimum observation fraction		0.8
Graph construction	Universe mode	Fixed replace (seed cohort then attrition replacement)	
Graph construction	Number of stocks		500
Graph construction	Selection rule	Market capitalization	
Graph construction	Eligibility update frequency	Each feature date under canonical fixed-replace mode	
Graph construction	Dependence measure	Spearman rank correlation	
Graph construction	Top-k neighbors		10
Graph construction	Graph symmetrization	Yes	
Graph construction	Node features	Rolling mean return; rolling volatility of return	
Graph construction	Forecast target	log_rv_fwd_30	
Dataset and splits	Training sample end		31/12/2014
Dataset and splits	Validation sample end		31/12/2018
Dataset and splits	Test sample end		31/12/2024
Dataset and splits	Feature scaler	Train-only standardization (zero mean and unit variance)	
Model architecture	Model type	GraphSAGE	
Model architecture	Number of GraphSAGE layers		4
Model architecture	Hidden dimension		256
Model architecture	Pooling method	Mean	
Training	Random seed		32
Training	Maximum epochs		100
Training	Learning rate		0.000005
Training	Early stopping patience		10
Training	Batch size		32
Training	Loss function	Mean squared error (log target)	
Evaluation and robustness	Evaluation splits	Training; Validation; Test	
Evaluation and robustness	H4 exclusion year(s)		2020
Evaluation and robustness	H4 stress window	2022-01-01-2022-10-31	
Evaluation and robustness	Loss functions used in formal benchmark comparison	Variance-based QLIKE and MSE (log target)	
Evaluation and robustness	Placebo ablation enabled	Yes	
Evaluation and robustness	Placebo edge seed		7

Table B.1. Canonical empirical configuration used in the thesis pipeline

Note: Monthly rebalancing exists as an alternative universe-construction mode in the pipeline, but the canonical specification uses fixed-replace universe maintenance with eligibility updated at each feature date.

H	Empirical question	procedure	Key comparison or statistic	Evaluation sample
H1	Whether the GNN forecast contains predictive information about future 30-trading-day realized market volatility.	Mincer-Zarnowitz predictive regression	Significance of the GNN slope coefficient	full test sample
H2	Whether the GNN achieves lower out-of-sample forecast loss than the VIX benchmark.	Diebold-Mariano-type loss comparison	Variance-based QLIKE and MSE (log target) loss differential	full test sample
H3	Whether the GNN adds predictive information beyond the VIX benchmark.	Incremental predictive regression	Significance of the GNN coefficient conditional on VIX	full test sample
H4	Whether the GNN retains its relative forecasting advantage once pre-specified extreme periods are excluded.	Restricted-subsample Diebold-Mariano and incremental regressions	Same H2 and H3 comparisons repeated on restricted subsamples	Test excluding 2020; test excluding stress window; test excluding both.

Table B.2. Mapping from empirical hypotheses to evaluation procedures in the canonical thesis pipeline.

7.3 Appendix C – Formal Evaluation and Hypotheses

Subsample	Role in formal evaluation	Definition
Full test	Baseline hold-out for formal tests	All trading days assigned to the test split in the canonical split definition. The configured end date for the test period is 2024-12-31.
Test excluding 2020	H4 robustness (calendar-year exclusion)	Test days excluding calendar year 2020. Inference on this subsample assesses sensitivity to dropping that calendar year.
Test excluding stress window	H4 robustness (stress-window exclusion)	Test days outside the inclusive stress window from 2022-01-01 through 2022-10-31. Inference on this subsample assesses sensitivity to excluding that episode.
Test excluding 2020 and stress window	H4 robustness (combined exclusions)	Test days remaining after excluding both calendar year 2020 and the stress window from 2022-01-01 through 2022-10-31. This applies both restrictions together on the hold-out test split.

Table C.1. Formal test subsamples used in robustness analysis (H4)

Hypothesis	Sample	Procedure	Loss / coefficient	Statistic	Primary p-value	Inference
H1	Full test	Mincer-Zarnowitz	GNN slope	8.8067	< 0.001	Very strong evidence in favor
H2	Full test	Diebold-Mariano	QLIKE	-1.0846	0.8609	No evidence in favor on the QLIKE criterion
H2	Full test	Diebold-Mariano	MSE(log)	1.9236	0.0272	Evidence in favor on the log-scale loss criterion
H3	Full test	Incremental regression	GNN slope conditional on VIX	1.9456	0.0517	Borderline evidence

Table C.2. Full-test formal hypothesis results (H1–H3)

H	Sample	Procedure	Loss / coefficient	Statistic	Primary p-value	Inference
H4	Test excluding 2020	Diebold-Mariano	QLIKE	0.1163	0.4537	No clear evidence in favor
H4	Test excluding 2020	Diebold-Mariano	MSE(log)	2.6236	0.0044	Very strong evidence in favor
H4	Test excluding 2020	Incremental regression	GNN slope conditional on VIX	1.5434	0.1227	No clear evidence in favor
H4	Test excluding stress window	Diebold-Mariano	QLIKE	-0.662	0.746	No clear evidence in favor
H4	Test excluding stress window	Diebold-Mariano	MSE(log)	3.1013	< 0.001	Very strong evidence in favor
H4	Test excluding stress window	Incremental regression	GNN slope conditional on VIX	1.9884	0.0468	Evidence of incremental information at the 5% level
H4	Test excluding 2020 and stress window	Diebold-Mariano	QLIKE	3.6119	< 0.001	Very strong evidence in favor
H4	Test excluding 2020 and stress window	Diebold-Mariano	MSE(log)	5.4644	< 0.001	Very strong evidence in favor
H4	Test excluding 2020 and stress window	Incremental regression	GNN slope conditional on VIX	0.5346	0.5929	No clear evidence in favor

Table C.3. Restricted-subsample robustness results (H4)

7.4 Appendix D – Supplementary Results

Model	Forecast variance in log space	Correlation with realized log-volatility	MZ intercept	MZ slope
GNN	0.0623	0.5399	0.1454	0.9719
VIX	0.1035	0.5993	0.2233	0.8371
HAR	0.1022	0.5151	0.7362	0.7239
Placebo	0.0667	0.5369	0.2365	0.9339

Table D.1. Calibration diagnostics for the full test sample

Model	Forecast variance in log space	Correlation with realized log-volatility	MZ intercept	MZ slope
GNN	0.0316	0.5539	-0.2716	1.1159
VIX	0.0669	0.659	-0.0225	0.9122
HAR	0.0657	0.5685	0.5068	0.794
Placebo	0.0357	0.554	-0.1082	1.0501

Table D.2. Calibration diagnostics for the test sample excluding 2020

Target-volatility quintile	GNN	VIX	HAR	Placebo
Q1 (lowest)	0.1213	0.3328	0.1702	0.1252
Q2	0.0393	0.1848	0.073	0.0431
Q3	0.034	0.1397	0.0494	0.0385
Q4	0.0997	0.1044	0.1017	0.0972
Q5 (highest)	0.4466	0.2355	0.3873	0.435

Table D.3. Mean squared forecast error in log space by target-volatility quintile on the full test sample

Target-volatility quintile	GNN	VIX	HAR	Placebo
Q1 (lowest)	0.1273	0.3375	0.1717	0.1284
Q2	0.0271	0.1492	0.0545	0.0285
Q3	0.0247	0.1218	0.0365	0.0276
Q4	0.0545	0.0841	0.0713	0.0523
Q5 (highest)	0.2158	0.0544	0.1211	0.2092

Table D.4. Mean squared forecast error in log space by target-volatility quintile on the test sample excluding 2020

7.5 Appendix E – Alternative configurations

Appendix E reports alternative tuned configurations evaluated alongside the canonical 20-trading-day specification. These configurations vary multiple design choices simultaneously, including graph-construction window, dependence measure, top-k setting, architecture family, and training hyperparameters. They should therefore be interpreted as alternative specification bundles rather than one-variable sensitivity checks. A 63-day tuned specification remained broadly competitive but did not improve held-out summary metrics or formal benchmark

comparisons relative to the canonical model. A 252-day tuned specification materially underperformed and was not retained as the final thesis configuration.

component	setting	default_20d	middle1_63d	yearwindow_252d
Graph construction	Rolling window length (trading days)		20	63
Graph construction	Minimum observation fraction		0.8	0.75
Graph construction	Edge dependence measure	Spearman rank correlation	Pearson correlation	Spearman rank correlation
Graph construction	Top-k neighbors per node		10	15
GNN model	Architecture family	GraphSAGE	GraphSAGE	GAT
GNN model	Number of message-passing layers		4	2
GNN model	Hidden dimension		256	256
GNN training	Learning rate	5.00E-06	2.00E-05	1.00E-05
GNN training	Maximum epochs	100	150	200
GNN training	Early stopping patience (epochs)	10	20	25
GNN training	Post-hoc log calibration of forecasts	No	Yes	Yes
Evaluation	Placebo edge ablation enabled	Yes	Yes	No

Table E.1. Comparison of the canonical and alternative empirical configurations evaluated

config	window	sample	model	mse_log	mae_log	qlike	n_samples
20-day canonical	20d	full_test	gnn	0.1482	0.2831	7.0893	1480
20-day canonical	20d	full_test	vix	0.1994	0.3619	6.9088	1480
20-day canonical	20d	full_test	har	0.1563	0.2815	7.1283	1480
20-day canonical	20d	full_test	placebo	0.1478	0.2816	7.0824	1480
20-day canonical	20d	excl_2020	gnn	0.09	0.2438	6.4515	1227
20-day canonical	20d	excl_2020	vix	0.1495	0.326	6.4575	1227
20-day canonical	20d	excl_2020	har	0.0911	0.24	6.4256	1227
20-day canonical	20d	excl_2020	placebo	0.0893	0.2417	6.4487	1227
63-day alternative	63d	full_test	gnn	0.1633	0.2814	7.3968	1480
63-day alternative	63d	full_test	vix	0.1994	0.3619	6.9088	1480
63-day alternative	63d	full_test	har	0.1563	0.2815	7.1283	1480
63-day alternative	63d	full_test	placebo	0.1614	0.2792	7.3815	1480

63-day alternative	63d	excl_2020	gnn	0.0924	0.2428	6.464	1227
63-day alternative	63d	excl_2020	vix	0.1495	0.326	6.4575	1227
63-day alternative	63d	excl_2020	har	0.0911	0.24	6.4256	1227
63-day alternative	63d	excl_2020	placebo	0.0899	0.2392	6.447	1227
252-day alternative	252d	full_test	gnn	0.3264	0.4325	8.295	1480
252-day alternative	252d	full_test	vix	0.1994	0.3619	6.9088	1480
252-day alternative	252d	full_test	har	0.1563	0.2815	7.1283	1480
252-day alternative	252d	excl_2020	gnn	0.2025	0.3591	6.9277	1227
252-day alternative	252d	excl_2020	vix	0.1495	0.326	6.4575	1227
252-day alternative	252d	excl_2020	har	0.0911	0.24	6.4256	1227

Table E.2. Comparison of held-out forecast performance across the canonical and alternative configurations

config	H	sample	procedure	loss_or_term	statistic	p_value
20-day canonical	H1	Full test	Mincer- Zarnowitz	GNN forecast slope (Mincer- Zarnowitz)	8.8067	1.29E-18
20-day canonical	H2	Full test	Diebold- Mariano	QLIKE	-1.0846	0.8609
20-day canonical	H2	Full test	Diebold- Mariano	MSE(log)	1.9236	0.0272
20-day canonical	H3	Full test	Incremental regression	GNN coefficient conditional on VIX	1.9456	0.0517
20-day canonical	H4	Test excluding 2020	Diebold- Mariano	QLIKE	0.1163	0.4537
20-day canonical	H4	Test excluding 2020	Diebold- Mariano	MSE(log)	2.6236	0.0044
20-day canonical	H4	Test excluding 2020	Incremental regression	GNN coefficient conditional on VIX	1.5434	0.1227
20-day canonical	H4	Test excluding stress window	Diebold- Mariano	QLIKE	-0.662	0.746
20-day canonical	H4	Test excluding	Diebold- Mariano	MSE(log)	3.1013	0.001

		stress window				
20-day canonical	H4	Test excluding stress window	Incremental regression	GNN coefficient conditional on VIX	1.9884	0.0468
20-day canonical	H4	Test excluding 2020 and stress window	Diebold-Mariano	QLIKE	3.6119	0.0002
20-day canonical	H4	Test excluding 2020 and stress window	Diebold-Mariano	MSE(log)	5.4644	2.32E-08
20-day canonical	H4	Test excluding 2020 and stress window	Incremental regression	GNN coefficient conditional on VIX	0.5346	0.5929
63-day alternative	H1	Full test	Mincer-Zarnowitz	GNN forecast slope (Mincer-Zarnowitz)	7.3439	2.07E-13
63-day alternative	H2	Full test	Diebold-Mariano	QLIKE	-1.1769	0.8804
63-day alternative	H2	Full test	Diebold-Mariano	MSE(log)	0.9453	0.1722
63-day alternative	H3	Full test	Incremental regression	GNN coefficient conditional on VIX	1.0722	0.2836
63-day alternative	H4	Test excluding 2020	Diebold-Mariano	QLIKE	-0.1176	0.5468
63-day alternative	H4	Test excluding 2020	Diebold-Mariano	MSE(log)	2.3266	0.01
63-day alternative	H4	Test excluding 2020	Incremental regression	GNN coefficient conditional on VIX	1.7665	0.0773
63-day alternative	H4	Test excluding stress window	Diebold-Mariano	QLIKE	-0.9903	0.839
63-day alternative	H4	Test excluding stress window	Diebold-Mariano	MSE(log)	1.6701	0.0474
63-day alternative	H4	Test excluding	Incremental regression	GNN coefficient	1.0205	0.3075

63-day alternative	H4	stress window Test excluding 2020 and stress window	Diebold-Mariano	conditional on VIX QLIKE	3.0291	0.0012
63-day alternative	H4	Test excluding 2020 and stress window	Diebold-Mariano	MSE(log)	5.1115	1.60E-07
63-day alternative	H4	Test excluding 2020 and stress window	Incremental regression	GNN coefficient conditional on VIX	0.5455	0.5854
252-day alternative	H1	Full test	Mincer-Zarnowitz	GNN forecast slope (Mincer-Zarnowitz)	0.8519	0.3942
252-day alternative	H2	Full test	Diebold-Mariano	QLIKE	-2.1145	0.9828
252-day alternative	H2	Full test	Diebold-Mariano	MSE(log)	-1.9268	0.973
252-day alternative	H3	Full test	Incremental regression	GNN coefficient conditional on VIX	-0.9528	0.3407
252-day alternative	H4	Test excluding 2020	Diebold-Mariano	QLIKE	-2.618	0.9956
252-day alternative	H4	Test excluding 2020	Diebold-Mariano	MSE(log)	-1.0629	0.8561
252-day alternative	H4	Test excluding 2020	Incremental regression	GNN coefficient conditional on VIX	-2.0736	0.0381
252-day alternative	H4	Test excluding stress window	Diebold-Mariano	QLIKE	-1.5555	0.9401
252-day alternative	H4	Test excluding stress window	Diebold-Mariano	MSE(log)	-0.6248	0.7339
252-day alternative	H4	Test excluding stress window	Incremental regression	GNN coefficient conditional on VIX	-0.9692	0.3324
252-day alternative	H4	Test excluding	Diebold-Mariano	QLIKE	-0.289	0.6137

252-day alternative	H4	2020 and stress window Test excluding 2020 and stress window	Diebold- Mariano	MSE(log)	2.4409	0.0073
252-day alternative	H4	2020 and stress window Test excluding 2020 and stress window	Incremental regression	GNN coefficient conditional on VIX	-1.3882	0.1651

Table E.3. Comparison of formal hypothesis-test results across the canonical and alternative configurations

configuration	Window label	sample	model	Forecast variance log	Corr with target log	Mz intercept	Mz slope
20-day canonical	20d	full_test	gnn	0.0623	0.5399	0.1454	0.9719
20-day canonical	20d	full_test	vix	0.1035	0.5993	0.2233	0.8371
20-day canonical	20d	full_test	har	0.1022	0.5151	0.7362	0.7239
20-day canonical	20d	full_test	placebo	0.0667	0.5369	0.2365	0.9339
20-day canonical	20d	excl_2020	gnn	0.0316	0.5539	-0.2716	1.1159
20-day canonical	20d	excl_2020	vix	0.0669	0.659	-0.0225	0.9122
20-day canonical	20d	excl_2020	har	0.0657	0.5685	0.5068	0.794
20-day canonical	20d	excl_2020	placebo	0.0357	0.554	-0.1082	1.0501
63-day alternative	63d	full_test	gnn	0.0809	0.5152	0.5934	0.8139
63-day alternative	63d	full_test	vix	0.1035	0.5993	0.2233	0.8371
63-day alternative	63d	full_test	har	0.1022	0.5151	0.7362	0.7239
63-day alternative	63d	full_test	placebo	0.0875	0.5055	0.6889	0.7678
63-day alternative	63d	excl_2020	gnn	0.0405	0.5653	0.0588	1.0055
63-day alternative	63d	excl_2020	vix	0.0669	0.659	-0.0225	0.9122
63-day alternative	63d	excl_2020	har	0.0657	0.5685	0.5068	0.794
63-day alternative	63d	excl_2020	placebo	0.0459	0.5612	0.2055	0.9375

252-day alternative	252d	full_test	gnn	0.0127	0.106	1.7089	0.4227
252-day alternative	252d	full_test	vix	0.1035	0.5993	0.2233	0.8371
252-day alternative	252d	full_test	har	0.1022	0.5151	0.7362	0.7239
252-day alternative	252d	excl_2020	gnn	0.0129	0.0734	2.0725	0.2311
252-day alternative	252d	excl_2020	vix	0.0669	0.659	-0.0225	0.9122
252-day alternative	252d	excl_2020	har	0.0657	0.5685	0.5068	0.794

Table E.4. Comparison of calibration and smoothing diagnostics across the canonical and alternative configurations

Target volatility quintile	Default 20d gnn	middle1 63d gnn	Yearwindow 252d gnn	vix
q1	0.1213	0.0878	0.0526	0.3328
q2	0.0393	0.0299	0.029	0.1848
q3	0.034	0.0301	0.0862	0.1397
q4	0.0997	0.1287	0.3179	0.1044
q5	0.4466	0.5403	1.1462	0.2355

Table E.5. Comparison of mean squared forecast error in log space by target-volatility quintile across configurations on the full test sample.

8 References

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